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Gravity-induced distortion dynamics of molten pool and keyhole during non-horizontal dual laser beam bilateral synchronous welding for Ti-6Al-4V alloy T-joints

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Abstract

Dual laser beam bilateral synchronous welding (DLBSW) has been proven as an effective fabrication technique for Ti-6Al-4V alloy T-joints. Nevertheless, the pronounced impact of gravity deflection on fluid flow and keyhole stability in DLBSW of spatially complex curved structures must be thoroughly evaluated, given its consequential effects on porosity and fatigue life. In this paper, the thermal-fluid coupling models of Ti-6Al-4V alloy T-joints during horizontal DLBSW and Non-Horizontal DLBSW at the gravity deflection angle (θ) of 25° and 40° were respectively established. The spatiotemporal characteristics of fluid flow under diverse gravitational regimes and the dynamic

Outline

evolution of the keyhole were comprehensively delineated. Additionally, the fundamental mechanism governing gravity-driven distortion of the molten pool and the resultant destabilization of the orifice under the dual-beam coupling effect was uncovered. It is revealed that gravitational deflection induces a reconfiguration of liquid metal flow from the upper molten pool, leading to diminished thermal energy delivery to the anterior wall. This, in turn, restricts the advancement of the solid/liquid interface and exacerbates thermal asymmetry. Moreover, gravitational deflection enhances bubble nucleation, impedes their removal, and compromises the liquid's void-filling efficacy, culminating in porosity formation. These outcomes offer significant guidance for porosity suppression strategies and inform the refinement of DLBSW techniques for T-joint manufacturing.

Keywords

Molten pool; Keyhole; Keyhole-induced porosity; Dual laser beam bilateral synchronous welding; Ti-6Al-4V alloy

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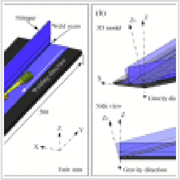
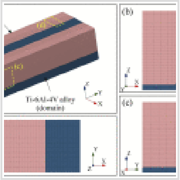
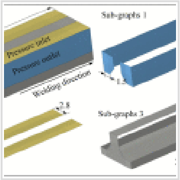
1. Introduction

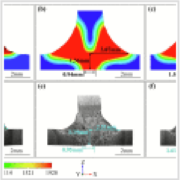
Driven by the demand for lightweight and high-performance development of large thin-walled structures, the connection efficiency and performance improvement of the skin-stringer T-joint structure under alternating loads have become one of the key directions that have attracted much attention in the advanced manufacturing field of complex lightweight alloy components for aircraft [1], [2]. The dual laser beam bilateral synchronous welding (DLBSW) technology can effectively ensure the integrity of the skin surface while significantly reducing the original structure weight, making it an ideal connection process for titanium alloy skin-stringer T-structure of aircraft [3], [4]. However, the interaction of the two laser beams generates a unified molten pool and a

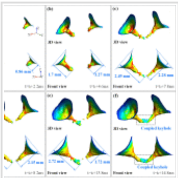
Figures (20)

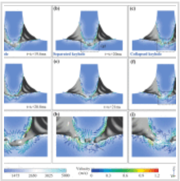












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Tables (3)

geometrically intricate coupled keyhole beneath the stringer [5]. Particularly in the context of large skin-stringer T-structure, the weld trajectory constitutes a complex spatial curve that varies in accordance with regional curvature. Under the influence of gravitational deflection, the morphology of both the molten pool and the keyhole undergoes significant deformation, thereby inducing pronounced disparities in the intensity of liquid convection, the dynamic evolution of the keyhole, the genesis of gas pores, and the resultant microstructural distribution [6]. Consequently, the attainment of continuous and stable welding along intricate spatial trajectories constitutes a formidable technical challenge.

Since Airbus Germany first proposed the DLBSW technology to manufacture the skin-stringer components, the main focus has been on optimizing the process of joining various light alloys [7], [8]. For the coplanar double lap-joint joined by DLBSW, Huang et al. [9] discussed the mechanism of deformation characteristics of the welded joints by establishing a thermal elastoplastic finite element model, and pointed out that the uneven distribution of lateral shrinkage is the primary cause for distortion. Han et al. [10] conducted DLBSW technology using different process parameters, and it was found that the grain coarsening and porosity defects in the HAZ greatly reduced the quality of the joints. Based on this method, the mechanism and influencing factors of the keyhole deep penetration effect in the DLBSW process have been revealed by Yan et al. [11]. Furthermore, by replacing the filler wire with a novel stringer element, the overall stability of the overall stability of the fabrication for complicated three-dimensional bending joints was effectively enhanced. This improvement leads to the attainment of T-joints with favorable weld appearance, particularly when a 0.8 mm×0.8 mm boss size was employed, as reported by Wu et al. [12]. Lv et al. studied systematically characterized the microstructure, phase transformation, and mechanical properties of titanium alloy [13]. Nevertheless, the mechanical characteristics of joints are compromised due to the porosity that is passed down from the joining technique used. Especially for the porosity caused by keyhole fluctuations during laser welding, which has been widely considered as the main defect in reducing fatigue strength.

According to Gouret et al. [14], the occurrence of porosity within titanium welds is ascribed to the existence of gas-forming materials, such as oil, grease, and moisture, on the surface of the joints. Therefore, a hydrogen dispersion-operated bubble growing mode was suggested in order to illustrate the formation mechanism of porosity during the joining of titanium alloys, as introduced by Huang et al. [15]. Meanwhile, Pang et al. [16] outlined that the swift cooling of the keyhole terminal surface prior to keyhole closure may result in a significant reduction in vapor pressure within the keyhole tip, because of the masking effect of the bulges. Xia et al. proposed the the dual-laser beam oscillating bilateral synchronous welding (DLBOW) process, and successfully reduced the porosity in T-joints [5]. For the special T-joint, the flow behavior and keyhole characteristics of the molten pool were more complex and a great deal of porosities formed in the joints was brought about by the instability of the welding process [17]. The collapse of the keyhole, precipitated by intense laser-material interactions, has been identified as a primary factor contributing to the formation of porosities [18].

Numerical modeling has become a popular way to investigate the instability of keyholes and keyhole-induced porosity, with providing a mutually beneficial alternative to traditional experimental methods [19], [20]. Recently, computational fluid dynamics (CFD) techniques have been utilized to investigate the formation process of keyhole and determined the most effective scanning techniques [21]. Meanwhile, the expansion and oscillation of the keyhole and the generation and mitigation of the keyhole-induced porosity, are highly dynamic processes involving extremely complex physics. In order to acquire a thorough comprehension of these processes, it is imperative to carry out simulation analysis utilizing the CFD model.

However, the complex three-dimensional spatially curved welds were conducted in practical applications. Assuming that the welding direction is considered constant on the horizontal plane, it is equivalent to the deflection of gravity during the actual welding process. Therefore, gravity has a crucial impact on the keyhole stability and the generation of keyhole-induced porosity. Gravity performs a crucial function in facilitating the convection of porosity flow, thereby enabling the gas phase to escape into the deep molten pool [22]. Adjusting the welding torch to reduce the effect of gravity on the molten pool flow could achieve stable keyholes [23]. The asymmetric flow of molten pool in the keyhole is the fundamental reason for the formation of porosity [24]. Some progress has been made in the above studies, but the impact mechanism of gravity deflection on the fluid flow of molten pool and keyhole stability in the DLBSW of T-joints remains unclear.

In this paper, the thermal-fluid coupling models of Ti-6Al-4V alloy T-joints during horizontal DLBSW and Non-Horizontal DLBSW at the gravity deflection angle of 25° and 40° were respectively established. The spatiotemporal characteristics of fluid flow under diverse gravitational regimes and the dynamic evolution of the keyhole were comprehensively delineated. Additionally, the fundamental mechanism governing gravity-driven distortion of the molten pool and the resultant destabilization of the orifice under the dual-beam coupling effect was uncovered. Besides, the bubble generation patterns in one cycle and the effect of gravity on the thermal fluid flow-induced porosity were studied. This study elucidates the mechanisms linking gravity-driven molten pool dynamics and keyhole stability to weld quality, and provides a scientific basis for optimizing the welding of spatially complex curved structures.

2. Experimental methods

The Ti-6Al-4V alloy in the fully-annealed state were selected as the base metal (BM) of T-joint, which was composed of a skin (500 mm × 90 mm) and a stringer (500 mm × 55 mm). The chemical components of Ti-6Al-4V alloy are listed in Table 1. To improve the stabilization of the welding process, bosses measuring 0.8 mm in both height and width were machined into the BM of each stringer utilizing a milling machine. To mitigate the risk of metallurgical porosity, a pre-weld surface preparation protocol, encompassing both mechanical and chemical cleaning, was mandatorily implemented.

Table 1. The chemical components of Ti-6Al-4V alloy (wt %).

Element	Fe	C	H	O	N	Al	V	Ti
TC4	≤0.3	≤0.1	≤0.015	≤0.2	≤0.05	5.5–6.8	3.5–4.5	Bal.

In the present research, the the impact of gravity on keyhole stability and the mitigation of keyhole-induced porosity during the DLBSW process were investigated. The schematic diagram of the DLBSW methodology is illustrated in Fig. 1. The experimental orientation employed to assess the influence of gravity is delineated in Fig. 1(b), where, *θ* represents the gravity deflection angle, i.e., the angle between the Y_0-axis and Y-axis. To study the influence of gravity on keyhole evolution and porosity development, the welding experiments were divided into 3 groups, i.e., θ=0°, θ=25° and θ=40°.

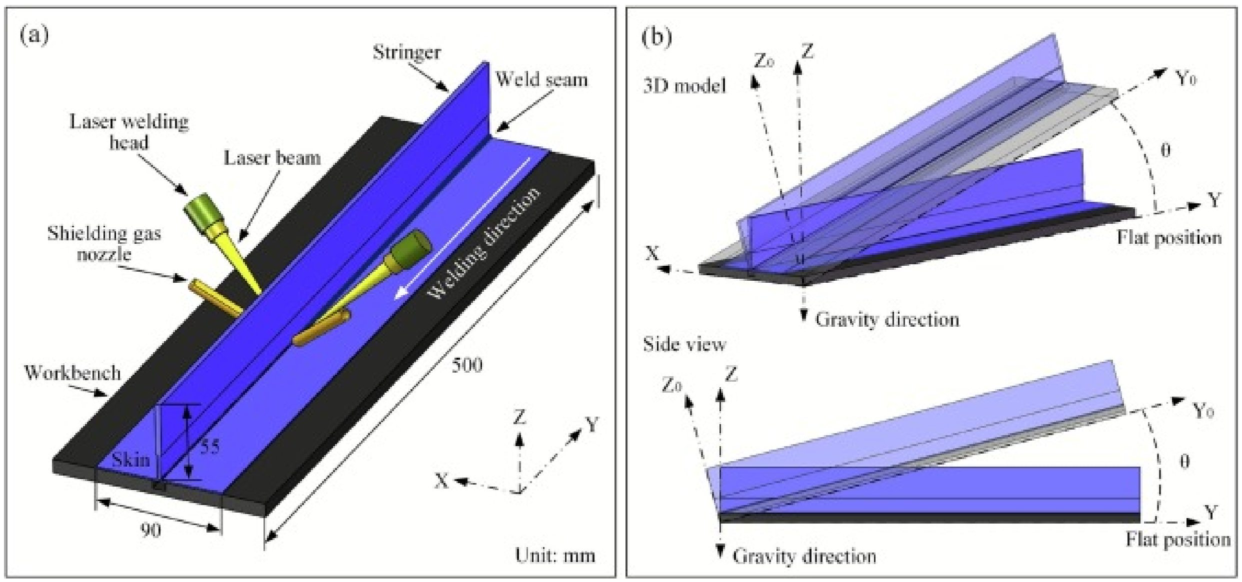

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Fig. 1. Schematic of DLBSW: (a) the experiment system; (b) definition of the welding position.

The findings of the welding at θ=0° were used as a reference and the results of the welding at θ=25° and θ=40° were utilized to study the mechanism of gravitational effects. The welding was carried out at a laser power of 2.2 kW with a defocusing distance of 0 mm. The welding speeds are 0.018 $m·s^{-1}$, 0.022 $m·s^{-1}$, and 0.028 $m·s^{-1}$, respectively. To enhance the quality of the weld seam, a pure argon protective gas was used, and it was blown into the molten pool with a flow rate of 25 $L·min^{-1}$. Table 2 provides a comprehensive overview of the welding parameters utilized across various gravity deflection states.

Table 2. DLBSW parameters for joining Ti-6Al-4V alloys T-joints.

No.	Gravity deflection angle (θ/°)	Laser power (P/kW)	Welding speed ($v/m·s^{-1}$)	Incident beam angle (α/°)
1.1	0	2.2	0.028	40
1.2			0.022	
1.3			0.018	
2.1	25	2.2	0.028	40
2.2			0.022	
2.3			0.018	
3.1	40	2.2	0.028	40
3.2			0.022	
3.3			0.018	

The cross-section weld specimens measuring 10 mm × 10 mm × 8 mm underwent mechanical cutting, grinding, cross-section polishing, and then etching using a Kroll etchant (10 ml HF, 100 ml H_2O, and 35 ml HNO_3). The respective macro morphology for diverse welding regimes was observed employing a Zeiss Axio Imager M2m Optical microscope facility. Based on the macro morphology detection result, the weld width, weld depth, and additional weld width were measured through image processing tools. A laser spectroscopic confocal microscope (KathMatic KC-X1000) was used to scan and measure the three-dimensional surface morphology and analyze the roughness of Ti-6Al-4V alloy welds. Furthermore, to further investigate the three-dimensional spatial position and appearance of the pores, the industrial X-ray CT scanning test method was conducted to obtain good-quality images and accurate flaw distributions.

3. Mathematical model and numerical simulation

Indeed, the interplay among the coupled heat source, combined molten pool, and substrate in DLBSW is exceedingly intricate, making it challenging to integrate all the relevant physical variables into a mathematical framework. According to the works of Panwisawas et al. [25] and Lu et al.[26], certain suppositions are invoked to simplify the mathematical model:

(1) The liquid metal exhibits laminar flow, characterized by its incompressibility as an ideal Newtonian fluid.

(2) The BM exhibits isotropy, and its properties are only determined by temperature.

(3) The influence of shielding gas on the molten pool and keyhole as well as the chemical reaction between the phases is neglected.

3.1. Governing equations

The heat transfer and fluid flow associated with tracking the molten pool's free surface during the DLBSW are described by the application of the subsequent governing equations: energy, momentum, mass, and Volume of Fluid (VOF) equations. They are depicted in the Descartes coordinate provided below [27], [28].

Mass conservation equation:

$$\frac{\partial \mathrm{p}}{\partial t}+\frac{\partial(\rho \mathrm{u})}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho \mathrm{w})}{\partial y}=0 \tag{1}$$

where t represents the welding time, ρ denotes the fluid density, and the velocity components in the$x$,y, and z directions are represented by u, v, and w respectively.

Momentum continuity equation:

x direction

$$\frac{\partial(\rho \mathrm{u})}{\partial t}+\frac{\partial(\rho u u)}{\partial x}-\frac{\partial(\rho u u 0)}{\partial x}+\frac{\partial(\rho \mathrm{u} v)}{\partial y}+\frac{\partial(\rho \mathrm{u} w)}{\partial z} =-\frac{\partial p_l}{\partial x}+\frac{\partial}{\partial x}\left(\mu \frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu \frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu \frac{\partial u}{\partial z}\right)+S u \tag{2}$$

y direction:

$$\frac{\partial(\rho v)}{\partial t}+\frac{\partial(\rho v u)}{\partial x}-\frac{\partial(\rho v u 0)}{\partial x}+\frac{\partial(\rho v v)}{\partial y}+\frac{\partial(\rho v w)}{\partial z} =-\frac{\partial p_l}{\partial y}+\frac{\partial}{\partial x}\left(\mu \frac{\partial v}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu \frac{\partial v}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu \frac{\partial v}{\partial z}\right)+S v \tag{3}$$

z direction:

$$\frac{\partial(\rho w)}{\partial t}+\frac{\partial(\rho w u)}{\partial x}-\frac{\partial(\rho w u 0)}{\partial x}+\frac{\partial(\rho w v)}{\partial y}+\frac{\partial(\rho w w)}{\partial z} =-\frac{\partial p_l}{\partial z}+\frac{\partial}{\partial x}\left(\mu \frac{\partial w}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu \frac{\partial w}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu \frac{\partial w}{\partial z}\right)+S_w \tag{4}$$

The values of P_l, u_0, and μ correspond to the fluid pressure, welding speed, and viscosity, respectively. The source terms of momentum in the$x$,y, and z directions are S_u, S_v, and S_w, respectively.

Energy conservation equation:

$$\frac{\partial(\rho H)}{\partial t}+\frac{\partial(\rho u H)}{\partial x}-\frac{\partial(\rho u_0 H)}{\partial x}+\frac{\partial(\rho v H)}{\partial y}+\frac{\partial(\rho w H)}{\partial z} = \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right)+S_E \tag{5}$$

where k represents the thermal conductivity and H denotes the enthalpy of mixing. The energy source term, S_E, can be given using the below equation:

$$S_E = Q - m_v l_v - \left[\frac{\partial}{\partial t}(\rho\Delta H)+\frac{\partial}{\partial x}(\rho u\Delta H)+\frac{\partial}{\partial y}(\rho v\Delta H)+\frac{\partial}{\partial z}(\rho w\Delta H)\right]+\frac{\partial}{\partial x}(\rho u_0\Delta H) \tag{6}$$

The laser heat source is denoted by Q, the mass transfer is represented by m_v, the latent heat of evaporation is denoted by L_v, and the latent heat of phase transformation is indicated by ΔH.

VOF equation:

$$\frac{\partial F}{\partial t}+\nabla\cdot(v_l F)=0 \tag{7}$$

where v_l represents the fluid velocity, F is the volume fraction.

3.2. Driven forces

The primary forces that propel the molten pool include thermal buoyancy, recoil pressure, surface tension, and gravity. The calculation of the recoil pressure induced by evaporation during the DLBSW process is conducted according to Li et al. [29]:

$$P_r = AP_0\exp\left(L_m\frac{T-T_b}{RTT_b}\right) \tag{8}$$

where, P_r represents the saturated pressure, A is a numerical factor with a value of 0.54, P_0 represents the atmospheric pressure, R is the constant of ideal gas, T_b denotes the boiling temperature, and L_m denotes the latent heat of evaporation.

The surface tension imposed on the liquid–gas interface is related to temperature variation, which references the work of Cho et al. [30]:

$$F_s = [MICK][\delta_0 + A_\delta(T - T_m)] \tag{9}$$

where, F_s represents the surface tension, K represents the curvature of the free surface, A_δ refers to the temperature gradient of the surface tension, δ_0 and T_m denote the surface tension coefficient and melting point respectively.

The buoyancy resulting from temperature changes can be described on the basis of the Boussinesq approximation, as stated by Shi et al. [31]:

$$F_b = \rho g \beta (T - T_l) \tag{10}$$

where the density is denoted by ρ, the thermal expansion coefficient by β, the gravity acceleration by g, and the melting temperature by T_l.

3.3. Laser heat source

To accurately depict the characteristics of the laser penetration welding heat source in simulations and streamline the model, a hybrid heat source combining the Gaussian surface heat source and the rotating Gaussian body heat source has been widely employed due to its exceptional versatility. The relationship between these two sub-heat sources can be calculated below:

$$Q_\eta = Q_s + Q_v \tag{11}$$

where the laser power is denoted by Q, the laser absorption rate of the material is denoted by η, and the valid energies of the body heat source and surface heat source are denoted by Q_v and Q_s, respectively.

The energy distribution for the Gaussian surface heat source is presented in the following equation:

$$q_s(x,y)=\frac{\eta_s Q_s}{\pi r_s^2}\exp\left(-\frac{\alpha_r\left(x^2+y^2\right)}{r_s^2}\right) \tag{12}$$

where η_s and r_s denote the effectiveness factor and radius of the heat source, respectively, and α_r represents the correcting factor utilized to align the simulation results with the empirical findings.

The expression for the energy distribution equation of the rotating body heat source can be found in the research conducted by Liang et al. [32].

$$q_v(x,y,z)=\frac{9\eta_v Q_v}{\pi h r_s^2\left(1-e^{-3}\right)}\exp\left(-\frac{9\left(x^2+y^2\right)}{r_s^2\ln\left(\frac{h}{z}\right)}\right) \tag{13}$$

where the useful depth and effectiveness factor of the heat source are denoted by h and η_v, respectively.

Compared to the spatial location of the thermal source in single beam laser welding, the spatial characteristics of the DLBSW are more complex. Therefore, the laser beams are focused at the center point of the corresponding boss and the spatial coordinate systems of the heat sources are transformed in this paper to accurately describe the spatial attributes of the laser beams installed on either side of the stringer. According to Fig. 1, it is illustrated that the welding direction is aligned parallel to the Y-axis of the initial coordinate system. Eqs. (14) and (15) represent the left and right heat sources acquired by counterclockwise and clockwise rotation, respectively, which are given as:

$$\begin{cases} y_{Ql}=y_l \\ z_{Ql}=(x-x_l)\cos\alpha+(z-z_l)\sin\alpha \\ x_{Ql}=(x-x_l)\sin\alpha-(z-z_l)\cos\alpha \end{cases} \tag{14}$$

$$\begin{cases} y_{Qr}=y_r \\ z_{Qr}=(z-z_r)\sin\alpha-(x-x_r)\cos\alpha \\ x_{Qr}=(x-x_r)\sin\alpha+(z-z_r)\cos\alpha \end{cases} \tag{15}$$

Here$x$,y, z represent the co-ordinates of the initial coordinate system, x_l, x_r, y_l, y_r, z_l, z_r are the co-ordinates of the coordinate system obtained after moving and rotating in the initial coordinate system, x_{Ql}, y_{Ql}, z_{Ql}, x_{Qr}, y_{Qr}, z_{Qr} are the coordinates of the coordinate system obtained after moving and rotating.

3.4. Boundary conditions

The surfaces of T-joints are set as wall boundaries that encompass heat radiation and convection can be expressed as Eq. (16). Originally, the air domain is populated with Ar at atmospheric pressure and ambient temperature (300K).

$$k\frac{\partial T}{\partial \vec{n}} = -h_c T + T_{ref} - \varepsilon\sigma T^4 + T_{ref}^4 \tag{16}$$

where the normal vector, represented by $\vec{n}$; the coefficient of convective heat transfer, referred to as h_c; the emissivity of radiation from the surface, denoted by ε; the Stefan-Boltzmann constant, symbolized as σ; the workpiece temperature, indicated as T; and finally, the ambient temperature, denoted as T_{ref}.

For the keyhole's liquid–air free surface, by considering heat dissipation through convection and radiation as well as metallic evaporation, the energy balance:

$$k\frac{\partial T}{\partial \vec{n}} = q_{absorb} - h_c T + T_{ref} - \varepsilon\sigma T^4 + T_{ref}^4 - W_{evp} H_v \tag{17}$$

where q_{absord} represents the absorbed laser energy. The liquid metal's vaporization laten and tvaporization rate are denoted as H_v and W_{evp}, respectively.

3.5. Numerical implementations

The thermophysical properties of the materials utilized in the modeling are tabulated in Table 3. The schematic representation of the calculation domain and mesh are shown in Fig. 2. The dimensions of the calculation domain are devised as 18 mm×7.6 mm×4.5 mm. The core area is uniformly meshed into hexahedral cells with dimensions of 0.312 mm×0.083 mm×0.081 mm, and the cell size is progressively enlarged in the area away from the weld to resolve the contradiction between accuracy and efficiency in the calculation. The model consisted of a total of 401,778 nodes and 417,976 cells.

Table 3. Thermophysical properties of the materials utilized in the simulations.

Physical properties	**Ti-6Al-4V**
Surface tension ($N{\cdot}m^{-1}$)	1.4
Solidus temperature (K)	1878
Temperature coefficient of surface tension ($Nm^{-1}{\cdot}K^{-1}$)	-0.26×10^{-3}
Liquidus temperature (K)	1928
Atmospheric pressure (Nm^{-2})	101,300
Molar mass ($g{\cdot}mol^{-1}$)	446.07
Latent heat of fusion ($J{\cdot}kg^{-1}$)	3.65×10^{5}
Ideal gas constant ($JK^{-1}{\cdot}mol^{-1}$)	8.314
Dynamic viscosity of liquid metal (Pa·s)	3.25×10^{3}
Thermal expansion coefficient (K^{-1})	8×10^{-6}
Density of liquid metal ($kg{\cdot}m^{-3}$)	4420
Latent heat of vaporization ($J{\cdot}kg^{-1}$)	9.092×10^{6}
Evaporation temperature (K)	3133

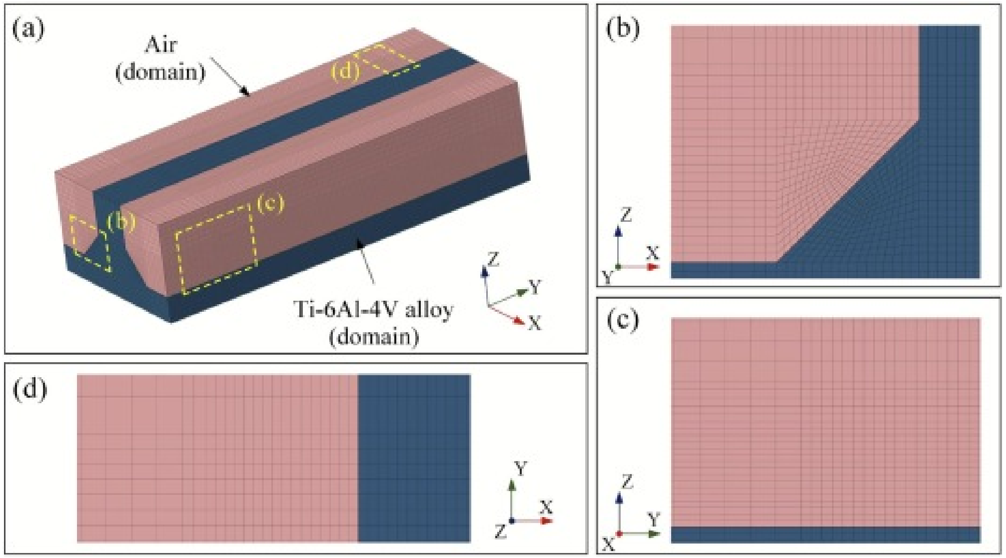

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Fig. 2. Schematic representation of the calculation domain and mesh: (a) 3D model; (b) enlarged view of the weld area; (c) enlarged view of the upper area; (d) enlarged view of the side area.

The simulation model included air domain and Ti-6Al-4V alloy domain, wherein all surfaces of the Ti-6Al-4V alloy domain were identified as wall boundaries exhibiting thermal convection and radiation. Furthermore, the upper surface of the air domain was designated as the pressure inlet, whereas the lateral surfaces were designated as the pressure outlet, as shown in Fig. 3. It should be noted that the filling gas in the air domain in the initial state is set as argon. The fluid dynamic equations were accurately solved utilizing the universal CFD software, ANSYS Fluent. By employing ANSYS Fluent, the numerical solution process accurately captures and analyzes the fluid dynamics within the mp. The introduction of energy and momentum sources was achieved through the implementation of user-defined functions (UDFs) coded in the C language. The coupling between pressure and velocity was implemented employing the Pressure-Implicit with

Splitting of Operators (PISO) algorithm. Specifically, the geometric modeling of the boss was reasonably simplified based on the actual macroscopic morphology of the weld.

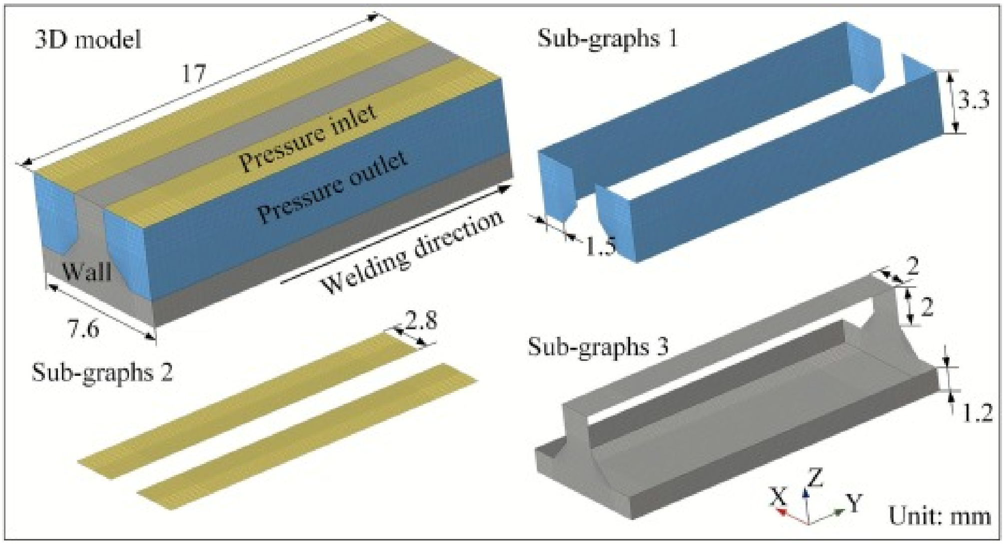

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Fig. 3. The boundary conditions used in the modeling.

4. Results and discussion

4.1. The dynamic evolution of the coupled keyhole during DLBSW

The CFD model provides a robust methodology for predicting the fluid dynamics within the molten pool, thereby enabling a comprehensive investigation into the coupled phenomena of keyhole stabilization and the mechanism of porosity formation under the influence of gravitational deflection. A comparative analysis of experimental and simulated results at various welding speeds for the 25° orientation are depicted in Fig. 4, where the characteristic values of the weld profile are quantitatively labeled. The maximum deviations between the experimental and simulated outcomes for the cross-sectional weld width, weld depth, and additional weld width at the 25° position are 0.56 mm, 0.06 mm, and 0.06 mm, respectively. Besides, the average deviations of weld width and weld depth are 0.497 mm and 0.037 mm, respectively. In conclusion, the concordance observed across all welding conditions substantiates the validity and reliability of the mathematical model developed in this study.

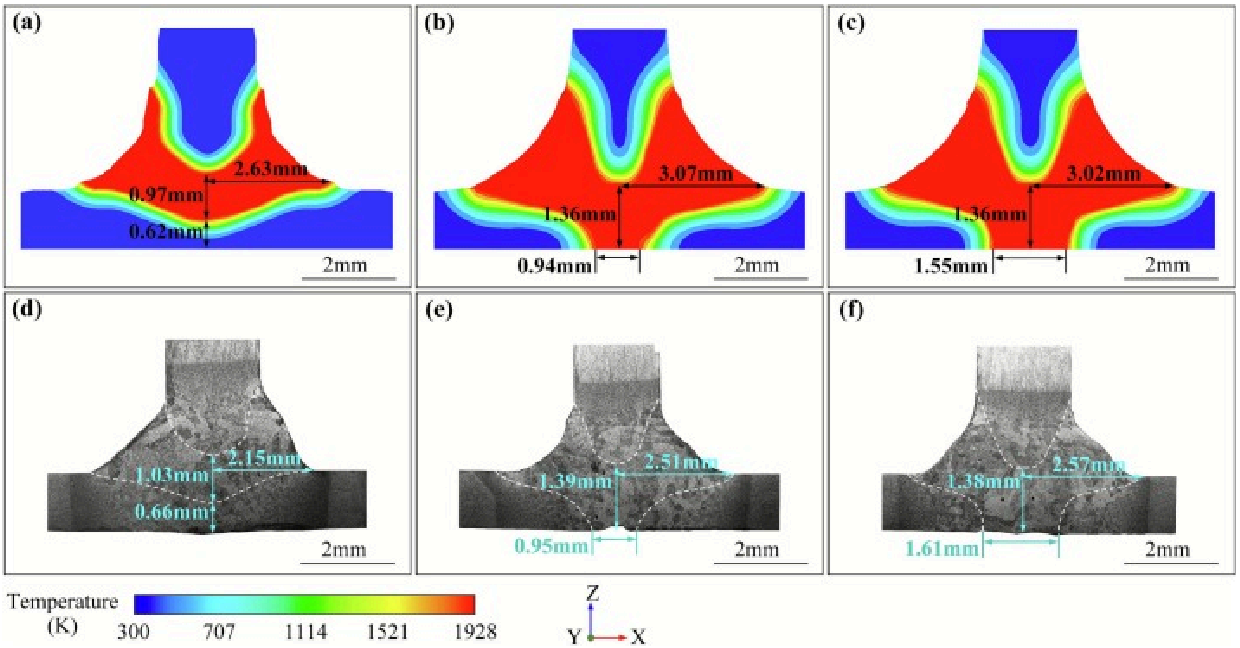

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Fig. 4. Comparison of experimental and simulated results at various welding speeds for the 25° orientation: (a)(d) 0.028 $m{\cdot}s^{-1}$; (b)(e) 0.022 $m{\cdot}s^{-1}$; (c)(f) 0.018 $m{\cdot}s^{-1}$.

Fig. 5 elucidates the simulated formation of a coupled keyhole under representative conditions (θ=0°, v=0.022 $m{\cdot}s^{-1}$), depicting the morphological characteristics of the keyhole walls from 3D view perspective and front view perspective. It should be noted that t_0 denotes the precise moment at which the laser beam initiates contact with the BM. Based on the morphological evolution of the coupled keyhole, the development can be delineated into three distinct stages. During the initial phase, as shown in Fig. 5(a-b), the recoil pressure generated by metallic vapor expels the liquid metal radially outward, thereby establishing a keyhole exhibiting pronounced bilateral symmetry. At $t=t_0+7.8$ ms, as shown in Fig. 5(c), the initial bilateral symmetry of the keyholes is disrupted, as evidenced by their divergent depths of 2.49 mm and 2.28 mm, respectively. Concurrently, the impingement of the vapor plume on the keyhole walls induces pronounced fluctuations at the keyhole root. Approximately 7 ms subsequent to this instability, the two discrete keyholes undergo coupling. During the formation of a coupled keyhole, numerous gauffers are generated on the keyhole's lower wall as the keyhole expands. In contrast, the upper wall exhibits a markedly smoother morphology, which signifies a more stable distribution of external pressure. The variation of external pressure can be attributed to multiple factors. A primary mechanism is the opposing action of vapor ejection and gravity on the upper liquid metal; the propulsive force from the vapor, directed antiparallel to gravity, partially counteracting the acquired momentum of the liquid metal. Conversely, the lower liquid metal encounters a propulsive force from the vapor ejects in the consistent direction of gravity, which leads to the differential manifestations of external pressure changes on different parts of the keyhole wall.

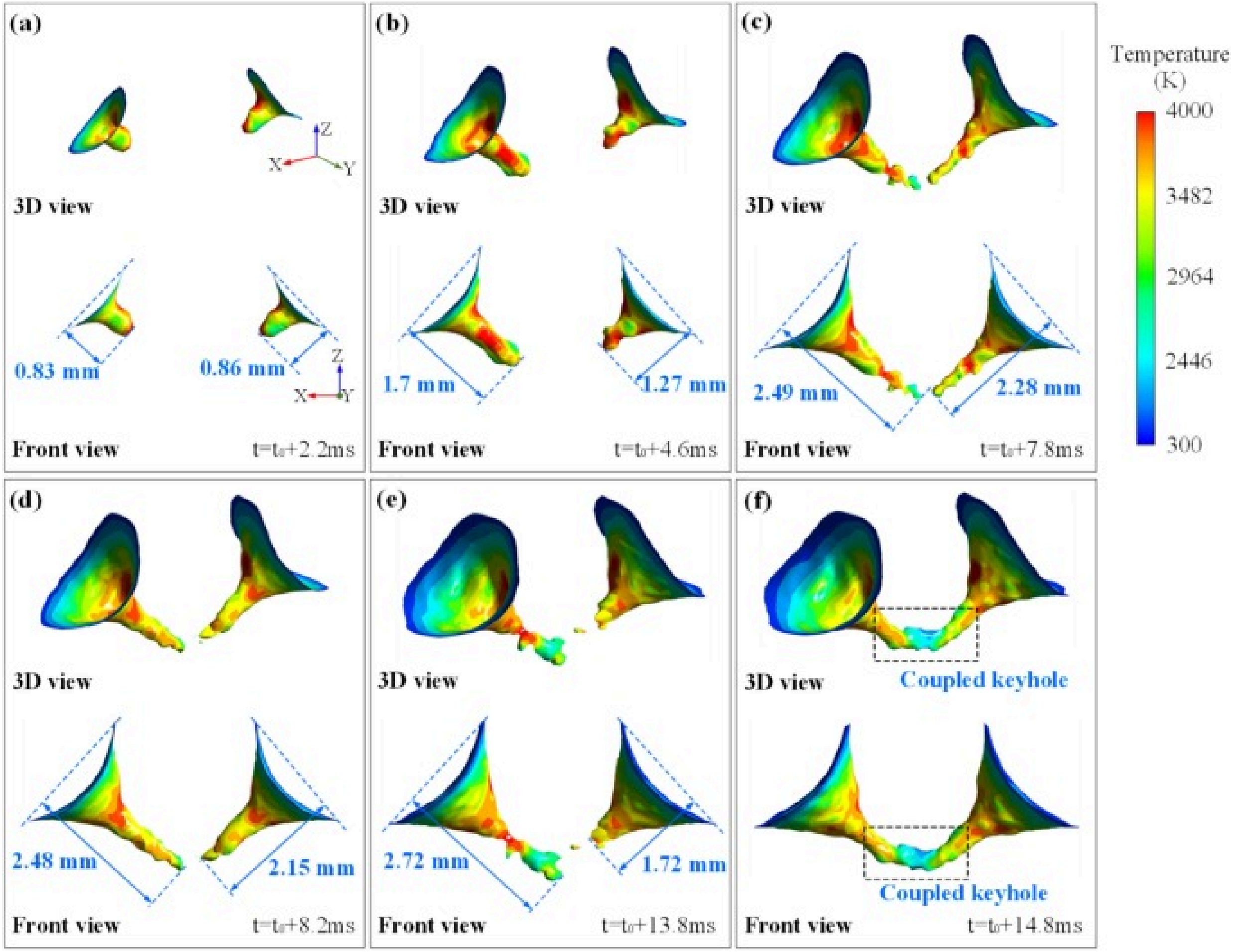

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Fig. 5. Comparison of keyhole morphology at different moments: (a) $t=t_0+2.2\,ms$; (b) $t=t_0+4.6\,ms$; (c) $t=t_0+7.8\,ms$; (d) $t=t_0+8.2\,ms$; (e) $t=t_0+13.8\,ms$; (f) $t=t_0+14.8\,ms$.

The evolution of the keyhole morphology from the coupled state to the separated state is depicted in Fig. 6. At $t=t_0+19.6\,ms$, the lower wall of the keyhole in the coupled region exhibits local swelling, resulting in the formation of bulges due to the synergistic effects of gravitational force and recoil pressure, as shown in Fig. 6(a). Additionally, the inclined keyhole hampers laser irradiation on the wall in the coupled region of the keyhole, thereby aggravating energy asymmetry and reducing surface tension efficacy. At $t=t_0+20.8\,ms$, the redistribution of liquid metal in the coupling, accompanied by the evolution of bottom vortices, culminates in the formation of a pronounced depression at the center of the keyhole, as depicted in Fig. 6(d). Two principal vortices are observed in the inferior region of the fully developed coherent molten pool. These vortices descend under the influence of gravitational force and subsequently converge in the central region, thereby generating vigorous convective flow. The vortex located at the lower center of the molten pool impinges upon the keyhole wall, and the resultant fluctuations in hydrodynamic pressure promote the contraction of keyhole and the formation of a depression, as illustrated in Fig. 6(b) and (g). During the transition of the keyhole from a coupled to a separated state, the upward velocity of melt flow, induced by the convergence of vortices in the bottom region of the molten pool, exhibits a transient increase followed by a gradual decay. After the liquid metal impacts the keyhole wall, the momentum and kinetic energy of the liquid metal induce significant morphological distortion of the keyhole, as depicted in Fig. 6(c)–(f). The instability in the middle region of the keyhole is primarily governed by the thermal convection of the molten pool, influenced by the combined effects of surface tension, recoil pressure, gravity, and hydrodynamic pressure. Consequently, the impingement of vortexes in the molten pool's bottom against the lower surface is the primary cause for the separation of the coupled keyhole.

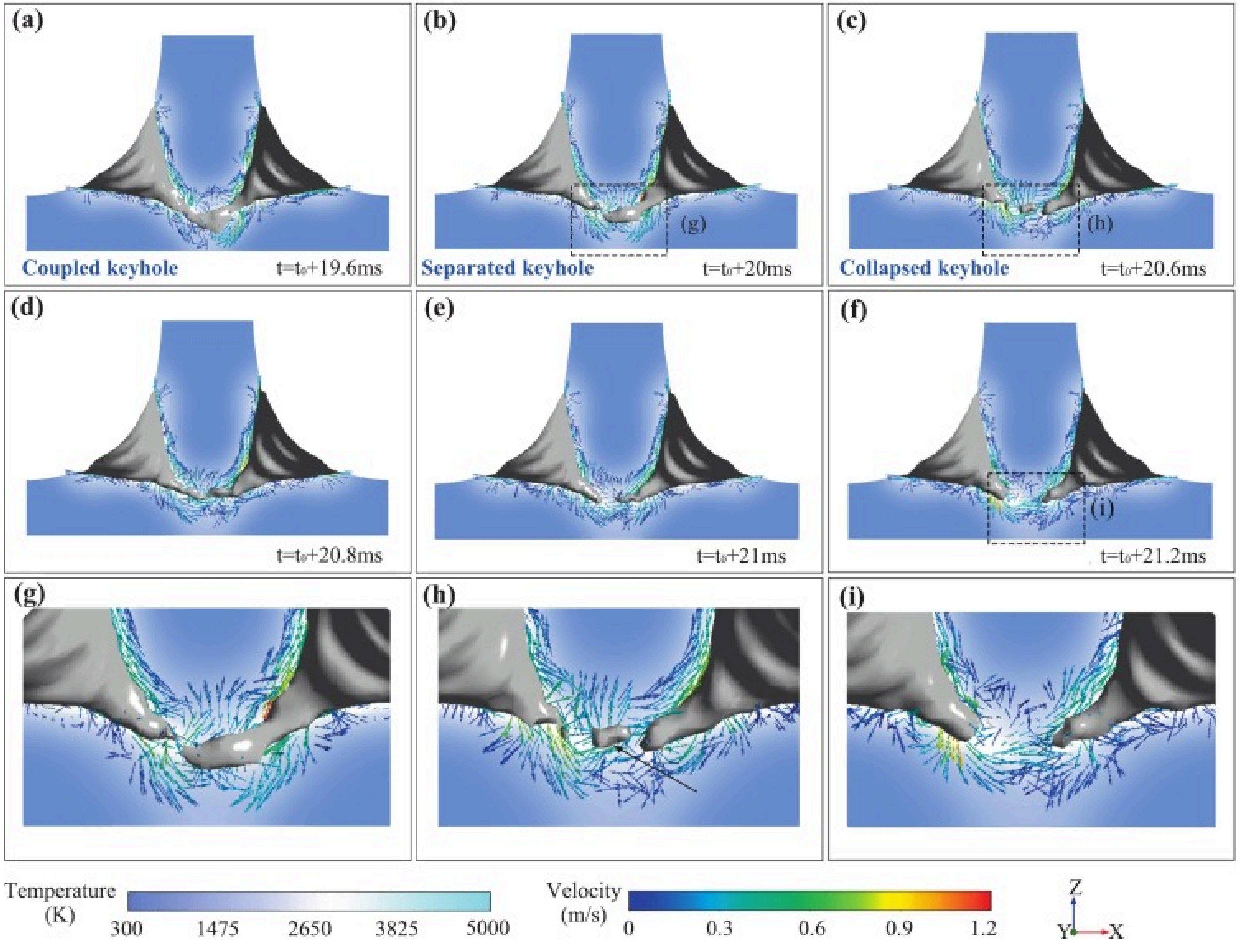

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Fig. 6. The evolution of the keyhole morphology from the coupled state to the separated state: (a) $t=t_0+19.6ms$; (b) $t=t_0+20ms$; (c) $t=t_0+20.6ms$; (d) $t=t_0+20.8ms$; (e) $t=t_0+21ms$; (f) $t=t_0+21.2ms$; (g~i) the magnified view in (b), (c), and (f).

The collective impingement of molten streams from multiple directions upon the central region of the keyhole induces localized depressions and oscillations along the lower wall, As illustrated in Fig. 7(a) and (b). These hydrodynamic perturbations, amplified by hydrokinetic pressure, ultimately precipitate the closure of the coupled keyhole. Furthermore, the coupled segment of the keyhole deviates from the incident laser path, while vapor-induced protrusions partially obstruct the incoming laser energy. This synergistic effect exacerbates the instability of the coupled region and accelerates its collapse. Eventually, the gas bubbles are generated upon the occlusion of the keyhole root. These bubbles become entrapped by the advancing solidification front before escape, ultimately resulting in the formation of porosity defects. It is noteworthy that bubble formation is not an irreversible process. As depicted in Fig. 6, at $t=t_0+20.6$ ms and $t=t_0+20.8$ ms, previously formed bubbles may be reabsorbed by the oscillating keyhole, thereby mitigating the onset of porosity.

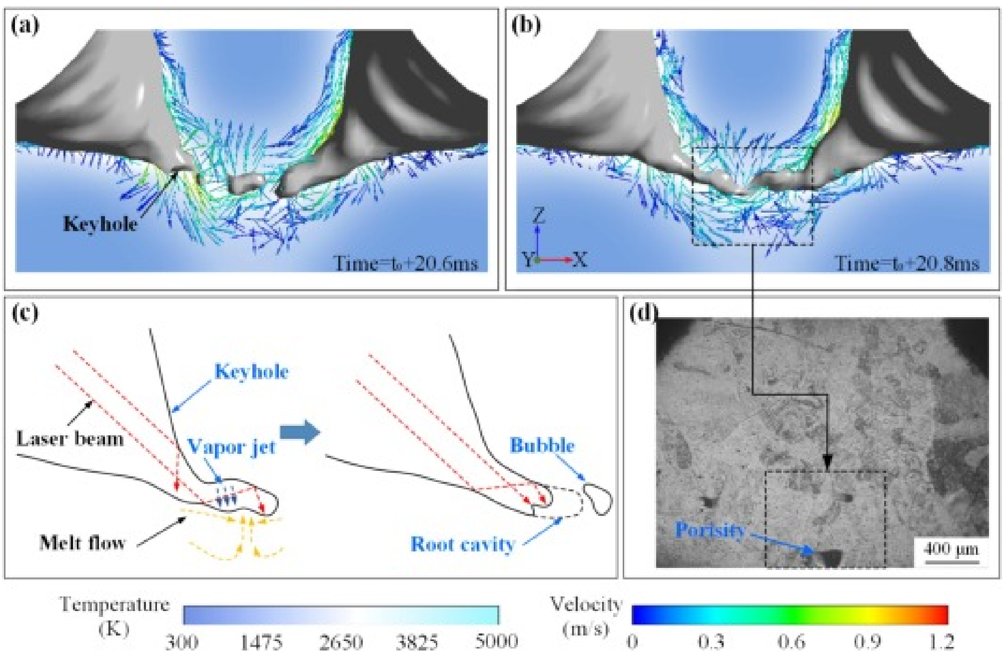

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Fig. 7. The formation mechanism of keyhole-induced porosity: (a) bubble formation; (b) bubble captured by the keyhole; (c) schematic diagram of keyhole closure; (d) keyhole-induced porosity.

4.2. Gravitationally-induced distortion behavior of molten pool during DLBSW

With the expectation of elucidate the influence of gravity on the molten pool flow behavior during non-horizontal DLBSW, a thermo-fluid coupling investigation was conducted at a laser power of 2200W, a welding speed of $0.022\,m{\cdot}s^{-1}$, and the varing θ of 0°, 25°, and 40°. These parameters ensured a fully penetrated molten pool configuration. Fig. 8 illustrates the longitudinal cross-sectional flow behavior of the molten pool during welding process under varying θ. In the coupled molten pool, the liquid metal flows along the keyhole's posterior wall in an upward direction driven by the Marangoni force and the recoil pressure, while simultaneously flowing along the solid/liquid interface in a downward direction influenced by gravity, culminating in the formation of a counterclockwise vortex. The vortices near the free surface are predominantly governed by surface tension characterized by a steep variation gradient. The maximum velocity at a θ of 0° is about $2.95\,m{\cdot}s^{-1}$, whereas the maximum velocity notably increases to approximately $3.27\,m{\cdot}s^{-1}$ at a θ of 25°. similarly, the maximum speed increases to $3.74\,m{\cdot}s^{-1}$ at the θ of 40°. It is observed that as the θ increases, the gravitational component in the y-direction synergistically interacts with the Marangoni force, collectively enhancing the backward flow of the surface fluid along the posterior wall of the keyhole. Concurrently, the hindrance of the z-direction gravity component on both recoil pressure and Marangoni force is attenuated, i.e., the counterclockwise flow of the fluid is facilitated.

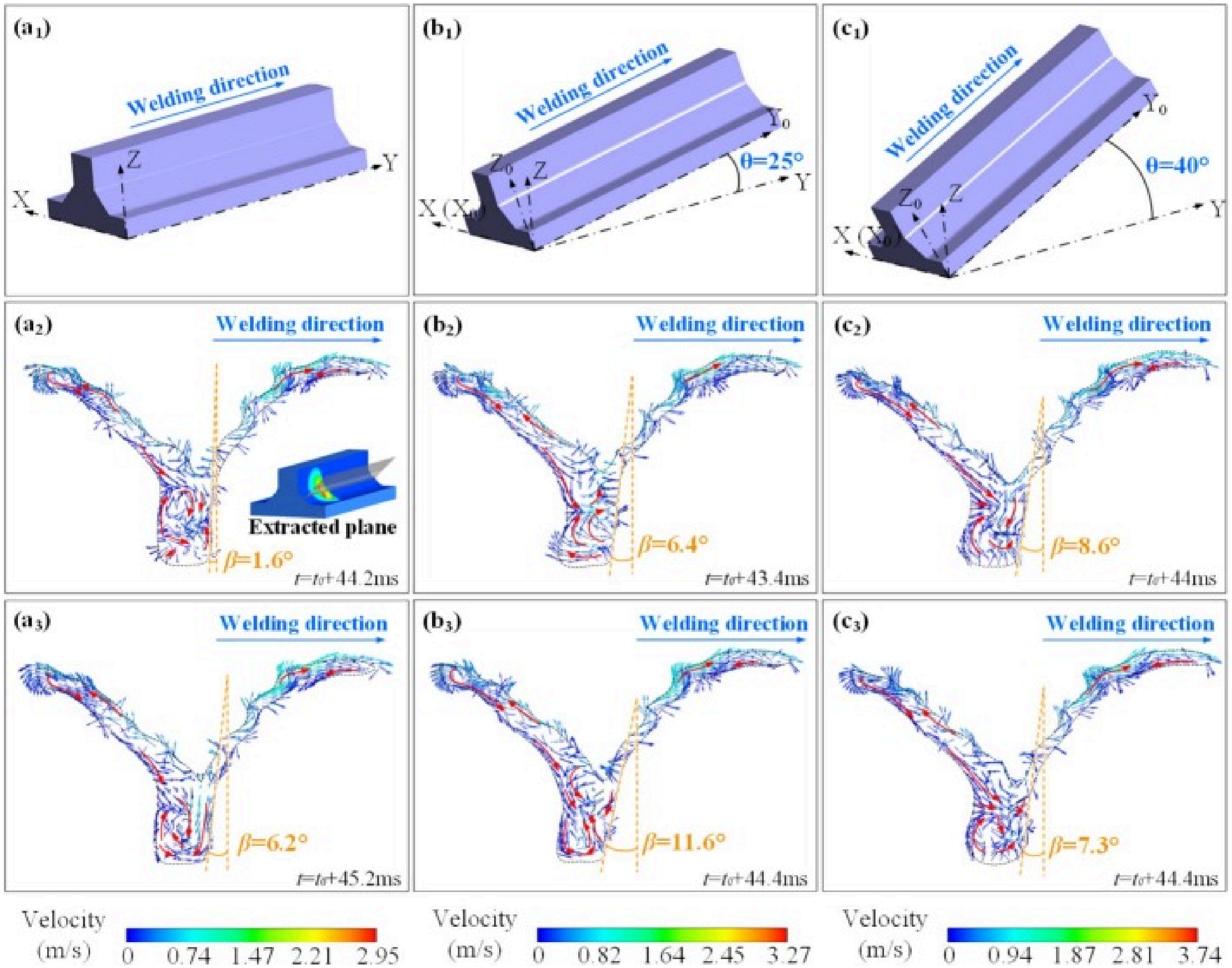

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Fig. 8. The longitudinal cross-sectional flow behavior of the molten pool during welding process under different θ: (a) θ=0°; (b) θ=25°; (c) θ=40°.

Under horizontal welding conditions, the molten fluid exhibits a characteristic clockwise vortex flow. As the θ progressively increases, the component of gravitational force parallel to the substrate surface enhances the lateral displacement of molten fluid descending from the upper region toward the posterior section of the pool. Concurrently, the attenuation of the gravitational component perpendicular to the substrate surface diminishes the impingement force exerted by the fluid directed toward the pool base. This disruption destabilizes the clockwise vortex characteristic of horizontal welding, inducing a transition toward disordered fluid motion in the lower region of the molten pool. Consequently, a proliferation of short, directionally unstable, and irregular flow patterns emerges, as illustrated in Fig. 8(a3), (b3), and (c3).

The weld surface morphology and roughness at different θ are displayed in Fig. 9. It is obseved that the surface roughness at both test positions for the weld with θ=0° (6.35 mm and 11.5 mm, respectively) is lower than those for the weld with θ=25° (22.42 mm and 21.38 mm, respectively) and the θ=40° (23.8 mm and 25.2 mm, respectively). Obviously, as the θ increases from 0° to 40°, the maximum velocity of the fluid in the molten pool elevates, accompanied by an increase in the surface roughness of the weld. Therefore, the increase in the θ reduces the stability of the upper part of the molten pool, which plays a decisive role on the appearance of the weld.

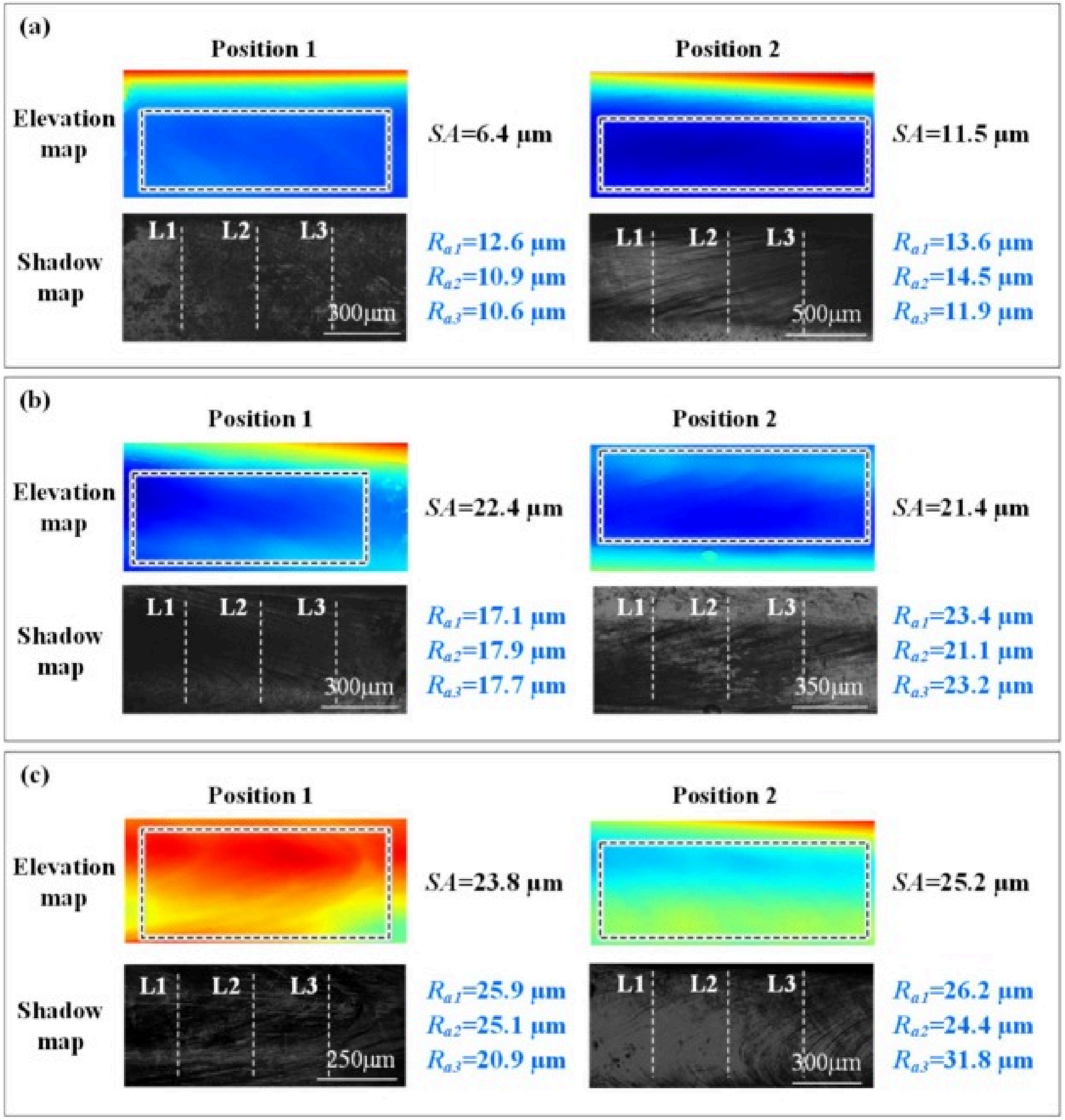

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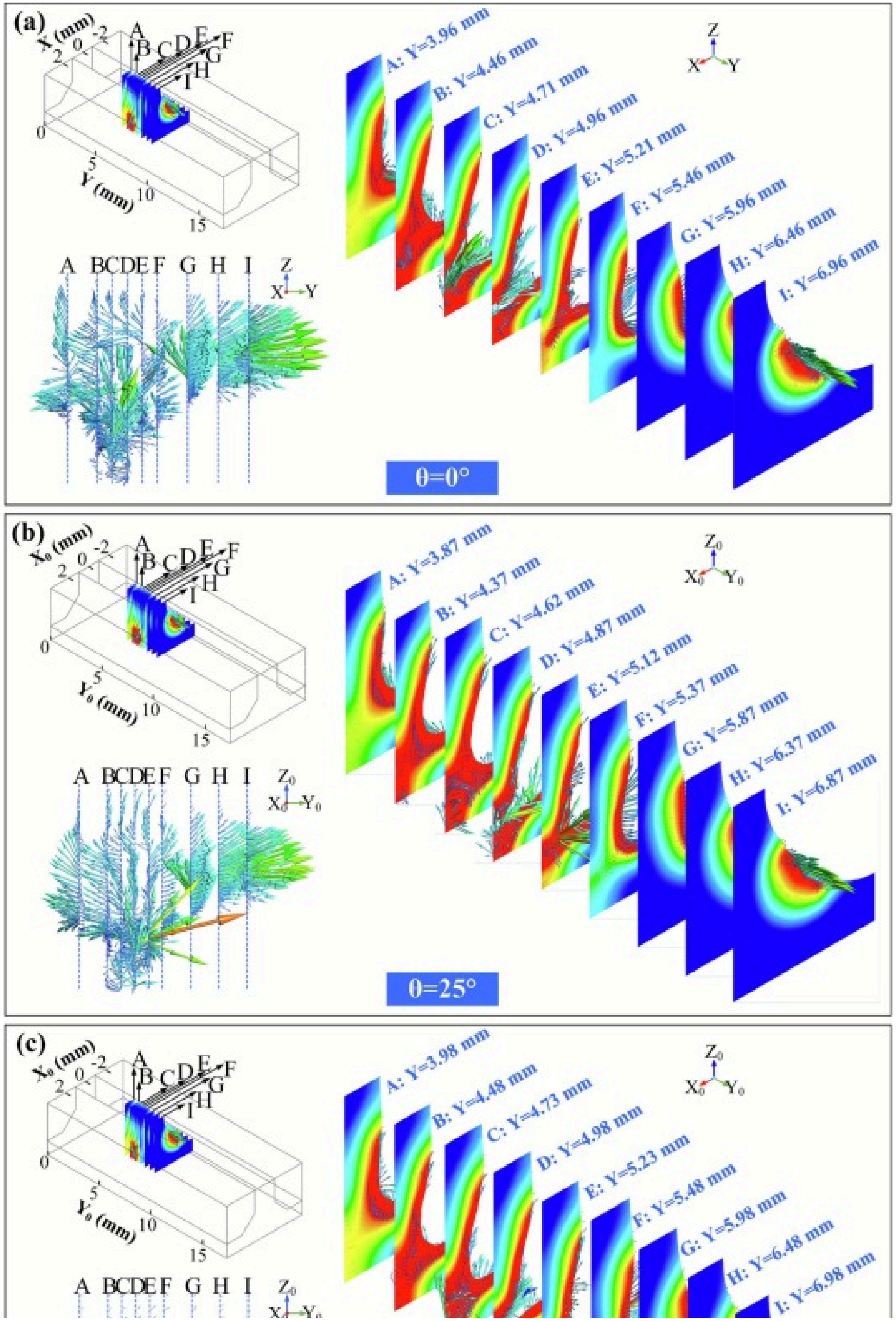
Fig. 9. The weld surface morphology and roughness at different θ: (a) θ=0°; (b) θ=25°; (c) θ=40°.

According to the fluid flow mechanism, during laser welding with θ=0°, the liquid metal from the molten pool's upper part first flows towards the anterior wall of the molten pool's lower part, and then towards the posterior wall of the molten pool. Differently, at the θ of 25° and 40°, the liquid metal from the molten pool's upper part directly flows towards the posterior wall and middle of the molten pool's lower part, which results in a marked reduction in thermal energy transferred to the anterior wall, thereby impeding the forward progression of the lower solid/liquid interface.

Furthermore, under the influence of a combination of multiple forces, the θ augments the flow of molten metal toward the posterior wall of the weld pool, thereby intensifying thermal energy transfer and promoting the formation of bulges. The evolution of vortices in the lower region of the molten pool is predominantly governed by hydrodynamic pressure, thermal buoyancy, and gravitational forces. Given that the keyhole bifurcates the molten pool into upper and lower sections, vigorous melt convection is induced along the keyhole periphery at the pool's center. In the absence of gravitational deflection, this convection exhibits a relatively coherent flow pattern, contributing to the maintenance of molten pool stability. However, when gravitational deviation occurs, the convection driven by hydrodynamic pressure, thermal buoyancy, and gravity is markedly enhanced. Consequently, the melt undergoes turbulent fluctuations and multidirectional eruptions, which impede effective heat transfer and dispersion, ultimately leading to localized heat accumulation and a reduction in molten pool stability.

Based on the VOF method, a three-dimensional instantaneous view of the thermal and velocity fields of the molten pool was simulated, as illustrated in Fig. 10. To enhance the observation of the molten pool contour, the maximum display temperature in the simulation was set to the 1928 K of the Ti-6Al-4V alloy. By customizing the maximum display temperature to the liquidus temperature, the simulation can more effectively capture and visualize the shape and behavior of the molten pool. Similarly, a more accurate comparison and analysis of the velocity field within the molten pool under different θ can be performed by setting the highest speed to 1.5 $m \cdot s^{-1}$. In order to observe the evolution of the molten pool, cross-sections are intercepted in the welding direction (y-direction) at intervals of 0.25 mm and 0.5 mm, with the red area indicating the molten pool. According to the resultant data obtained by calculating the fluid velocities at various cross-sections, the maximum velocities of the fluid within the molten pool and the respective speed components in the y and z directions are 0.95 $m \cdot s^{-1}$, 0.94 $m \cdot s^{-1}$, and 0.79 $m \cdot s^{-1}$ at the θ of 25°. The maximum velocities of the fluid within the molten pool and the respective speed components in the y and z directions are 0.91 $m \cdot s^{-1}$, 0.89 $m \cdot s^{-1}$, and 0.66 $m \cdot s^{-1}$ at the θ of 40°. By comparison, the corresponding maximum velocities of fluid flow within the molten pool are respectively 0.88 $m \cdot s^{-1}$, 0.88 $m \cdot s^{-1}$, and 0.77 $m \cdot s^{-1}$ at θ of 0°.


(a)
A: Y=3.96 mm
B: Y=4.46 mm
C: Y=4.71 mm
D: Y=4.96 mm
E: Y=5.21 mm
F: Y=5.46 mm
G: Y=5.96 mm
H: Y=6.46 mm
I: Y=6.96 mm
θ=0°
(b)
A: Y=3.87 mm
B: Y=4.37 mm
C: Y=4.62 mm
D: Y=4.87 mm
E: Y=5.12 mm
F: Y=5.37 mm
G: Y=5.87 mm
H: Y=6.37 mm
I: Y=6.87 mm
θ=25°
(c)
A: Y=3.98 mm
B: Y=4.48 mm
C: Y=4.73 mm
D: Y=4.98 mm
E: Y=5.23 mm
F: Y=5.48 mm
G: Y=5.98 mm
H: Y=6.48 mm
I: Y=6.98 mm

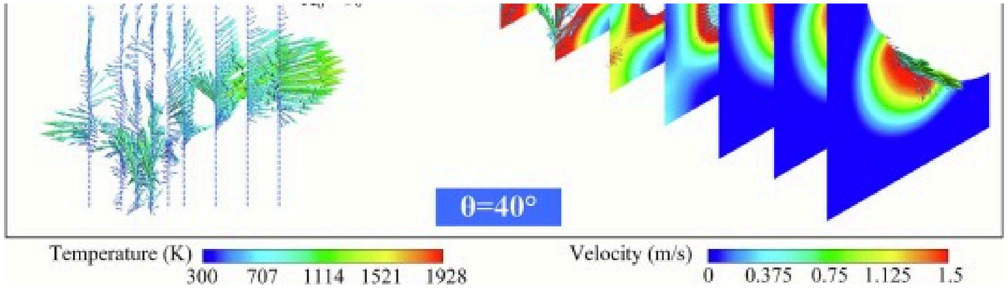


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Fig. 10. The thermal and velocity fields of the molten pool under different θ: (a) θ=0°; (b) θ=25°; (c) θ=40°.

This finding suggests that the flow velocity of the fluid increases significantly under large-angle gravity deflection conditions, exhibiting a higher degree of fluidity. Although the maximum flow velocity at a θ of 25° exceeds that observed under the current condition, this discrepancy does not contradict the previously drawn conclusion. This is attributable to the fact that the absolute value of the minimum velocity across various directions is markedly higher than that recorded at a θ of 40°. Consequently, the molten pool demonstrates improved fluidity alongside a marked escalation in disorder flow behavior.

Under gravity deflection conditions, the generation of a gravitational component in the y-direction and a reduction of the gravitational component in the z-direction affects the flow of liquid metal in the anterior of the molten pool. Specifically, the gravity component in the y-direction counteracts the action played by the recoil pressure and the Marangoni force on the liquid metal flow in the anterior of the molten pool, thus leading to a reduction in the velocity, as illustrated in Fig. 11. With θ=0°, it is reveralled that the molten metal flows backward without exhibiting reflux in the upper region. This behavior elucidates the observed reduction in molten pool stability and the corresponding increase in weld surface roughness under elevated θ. The comparative analysis of the directional flow velocities within the molten pool across varying θ reveals that the velocity-y is comparatively less susceptible to gravitational influence than velocity-z. Furthermore, the velocity-z exhibits a pronounced escalation with increasing θ. This effect is particularly conspicuous in the posterior region of the molten pool, which encompasses a larger molten pool area.

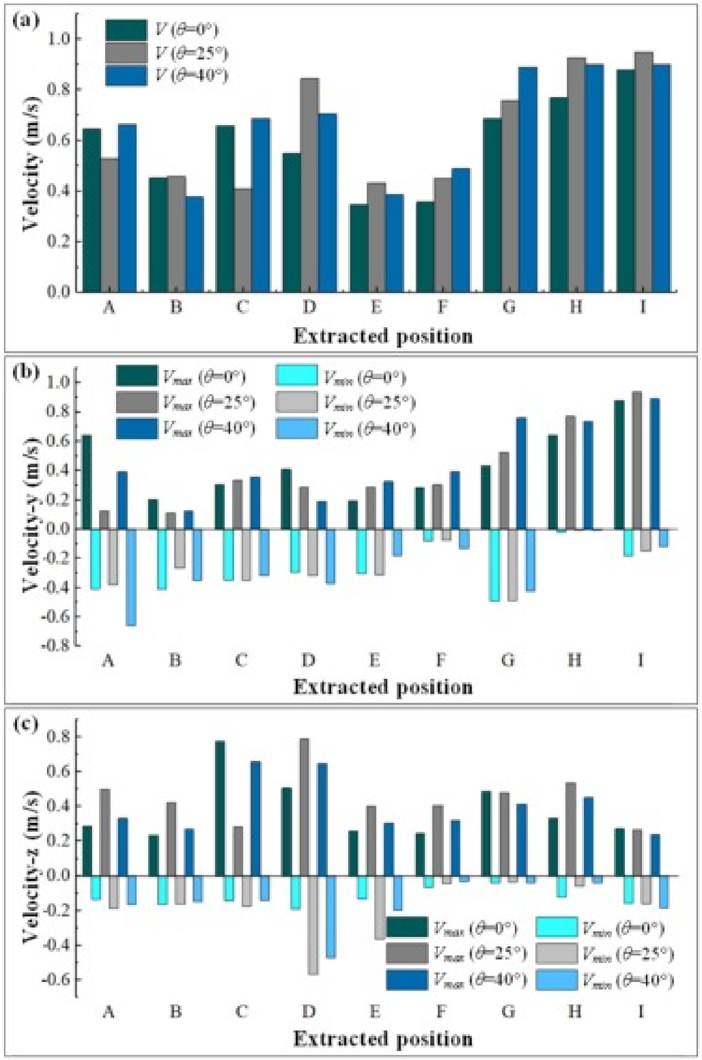

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Fig. 11. The fluid flow velocities in different directions within the molten pool under various θ: (a) velocity; (b) velocity-y; (c) velocity-z.

4.3. Gravitationally-driven dynamic behavior of keyhole during DLBSW

Fig. 12 presents the schematic representation the force distribution on the keyhole wall during DLBSW. Herein, the multiple local protrusions on the keyhole wall are simplified as a straight line. The angles formed by the front and rear keyhole walls with the laser incident path are defined as α_f and α_r, respectively, the magnitudes of which are primarily governed by the laser heat input. The gravitational force (G), can be decomposed into two components: a normal component, G_n, acting in the same direction as the recoil pressure, and a tangential component, G_t, perpendicular to the same direction of the recoil pressure. The G_t in conjunction with surface tension and gas/metal friction forces (F_f), drives the migration of bulges along the tangent to the keyhole wall. Conversely, the G_n combined with the recoil pressure, dominates the radial dynamics of the keyhole, governing its outward expansion or inward collapse. For horizontal welding, the gravitational components at varying positions can be calculated using the following equations:

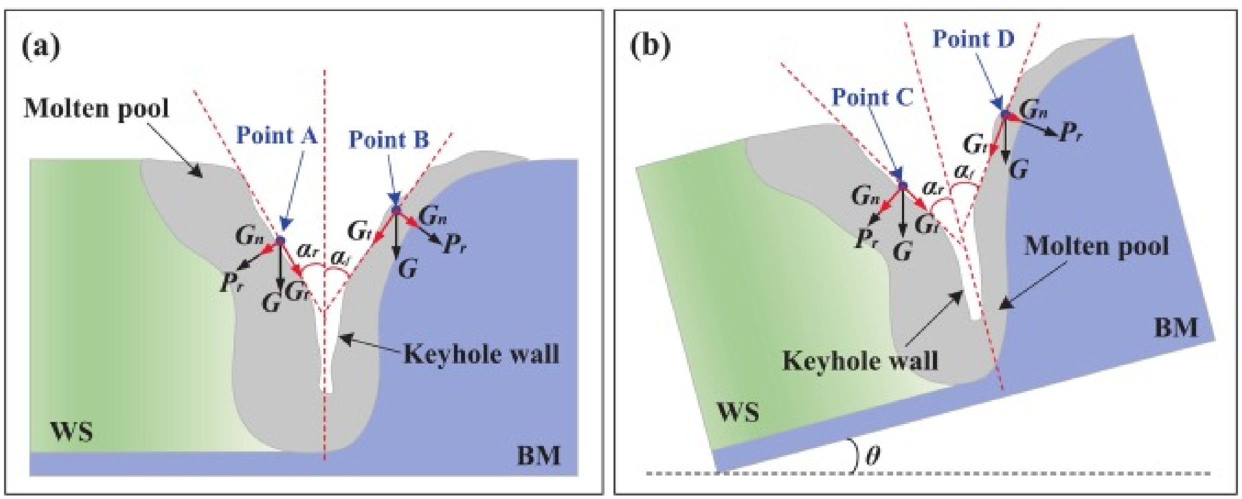

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Fig. 12. The force distribution on the keyhole wall during DLBSW: (a) horizontal welding (θ=0°); (b) non-horizontal welding.

Point A:

$$G_n = G \sin \alpha_r \tag{18}$$

$$G_t = G \cos \alpha_r \tag{19}$$

Point B

$$G_n = G \sin \alpha_f \tag{20}$$

$$G_t = G \cos \alpha_f \tag{21}$$

For non-horizontal welding, the gravitational components at varying positions can be calculated using the following equations:

Point C:

$$G_n = G \sin (\alpha_r + \theta) \tag{22}$$

$$G_t = G \cos (\alpha_r + \theta) \tag{23}$$

Point D:

$$G_n = G \sin (\alpha_f - \theta) \tag{24}$$

$$G_t = G \cos (\alpha_f - \theta) \tag{25}$$

Furthermore, under different gravitational effects, the main driving force for the expansion of the front and rear keyhole wall can be respectively obtained as:

$$F_{nf} = AP_0 \exp\left(L_m \frac{T-T_b}{RTT_b}\right) + G\sin\left(\alpha_f - \theta\right) \tag{26}$$

$$F_{nr} = AP_0 \exp\left(L_m \frac{T-T_b}{RTT_b}\right) + G\sin\left(\alpha_r + \theta\right) \tag{27}$$

The main driving force for the migration of bulges on the front and rear keyhole wall under different gravitational effects can be respectively obtained as:

$$F_{tf} = [MICK][\delta_0 + A_\delta(T - T_m)] + G\cos\left(\alpha_f - \theta\right) + F_f \tag{28}$$

$$F_{tr} = [MICK][\delta_0 + A_\delta(T - T_m)] + G\cos\left(\alpha_r + \theta\right) + F_f \tag{29}$$

It is revealed that, as the deflection angle θ increases, the gravitational component promoting keyhole expansion from the rear wall correspondingly increases, whereas the analogous component on the front wall decreases. Conversely, as the deflection angle θ increases, the tangential gravitational component driving bulge migration diminishes on the rear keyhole wall, whereas, increases on the front keyhole wall.

Fig. 13 presents a reasonable comparative analysis of numerical simulations depicting the dynamic behavior of keyhole under various gravity deflection conditions. Based on the morphological characteristics of the keyhole, the keyhole can be divided into two regions: the opening region and the coupling region. The open region is located on both sides of the keyhole, whereas the coupling region is situated in the middle of the molten pool. The classification of these two keyhole regions is essential for understanding its formation and stability. As indicated in Fig. 13, the anterior wall of the keyhole opening region is an inclined surface exhibiting a pronounced bulge, which migrates along free surface over time. As a result of the recoil pressure, the keyhole tends to open. However, various factors, i.e., surface tension, hydrodynamic pressure, and Marangoni forces, collectively inhibit the expansion of the keyhole. At the θ of 25° and 40°, the obliquity of the keyhole's front wall is greater compared to the flat position, which increases the contact area with the laser, enhancing energy absorption and exacerbating the uneven distribution of heat. Simultaneously, the liquid metal along the keyhole's anterior wall exhibits a relatively high flow velocity. This accelerated flow results in the impingement of the liquid metal against the keyhole wall, thereby disrupting the equilibrium among the forces acting upon the keyhole. Consequently, the non-uniform force distribution exerted on the anterior wall induces pronounced oscillations, rendering the keyhole highly unstable under gravity-deflected conditions.

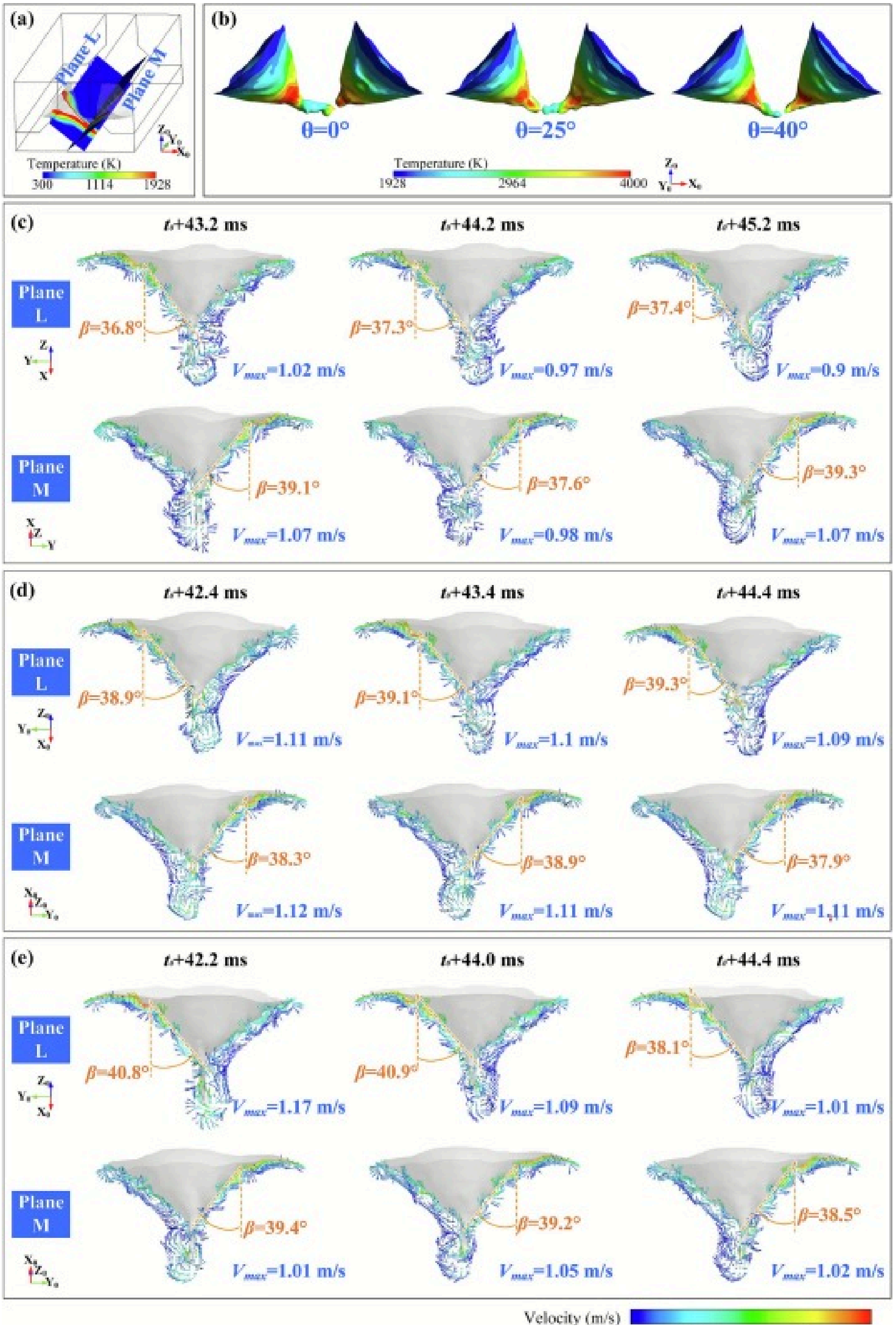
(a)
Plane L
Plane M
Temperature (K)
300
1114
1928
(b)
θ=0°
θ=25°
θ=40°
Temperature (K)
1928
2964
4000
(c)
t_s+43.2 ms
t_s+44.2 ms
t_s+45.2 ms
Plane L
β=36.8°
β=37.3°
β=37.4°
V_{max}=1.02 m/s
V_{max}=0.97 m/s
V_{max}=0.9 m/s
Plane M
β=39.1°
β=37.6°
β=39.3°
V_{max}=1.07 m/s
V_{max}=0.98 m/s
V_{max}=1.07 m/s
(d)
t_s+42.4 ms
t_s+43.4 ms
t_s+44.4 ms
Plane L
β=38.9°
β=39.1°
β=39.3°
V_{max}=1.11 m/s
V_{max}=1.1 m/s
V_{max}=1.09 m/s
Plane M
β=38.3°
β=38.9°
β=37.9°
V_{max}=1.12 m/s
V_{max}=1.11 m/s
V_{max}=1.11 m/s
(e)
t_s+42.2 ms
t_s+44.0 ms
t_s+44.4 ms
Plane L
β=40.8°
β=40.9°
β=38.1°
V_{max}=1.17 m/s
V_{max}=1.09 m/s
V_{max}=1.01 m/s
Plane M
β=39.4°
β=39.2°
β=38.5°
V_{max}=1.01 m/s
V_{max}=1.05 m/s
V_{max}=1.02 m/s
Velocity (m/s)

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Fig. 13. The the dynamic behavior of keyhole under various gravity deflection conditions: (a) the schematic diagram of extracted plane L and plane M; (b) Three-dimensional keyhole shape under various gravity deflection conditions; (c) keyhole dynamic behavior at θ=0°; (d) keyhole dynamic behavior at θ=25°; (e) keyhole dynamic behavior at θ=40°.

It is worth noting that the amplitude of such oscillations appears to be unimpressive when compared to the coupling region and the connection area between the coupling region and the opening region. As illustrated in Fig. 13, a distinct and abrupt turning occurs at the juncture of these two regions. This transition leads to the generation of a powerful vapor flow within the keyhole, impetuously impacting the lower surface and contributing to the forming of a bulge. Specifically, the bulges are moved downward along the keyhole wall by a combination of vapor flow and molten pool vortices. Furthermore, the turnings and bulges of the keyhole wall result in a drastic reduction of the absorption of laser energy in the coupling zone. This significantly impacts the evaporation state within the coupling region, leading to a lack of effective support for the keyhole wall from the interior.

Bedises, a quantitative study was conducted on the angle between the front keyhole wall and XOZ or X_0OZ_0 plane, which is difined as β. As illustrated in Fig. 13(c), during horizontal DLBSW, the δ of the left keyhole, measured at successive time intervals, are recorded as 36.8°, 37.3°, and 37.4°. In contrast, the β for the right keyhole are determined to be 39.1°, 37.6°, and 39.3°, respectively. As illustrated in Fig. 13(e), during non-horizontal DLBSW at the θ of 40°, the β of the left keyhole, measured at successive time intervals, are recorded as 40.8°, 40.9°, and 38.1°. while the δ for the right keyhole are determined to be 39.4°, 39.2°and 38.5°, respectively. A comparative analysis reveals that the β values of the keyhole front wall are systematically augmented as the θ increases. At θ=25°, the mean β values for the right un-contracted and left contracted walls are elevated by 1.2° and 0.43°, respectively, relative to the horizontal DLBSW. At θ=40°, the left un-contracted and right contracted walls exhibit increments of 2.76° and 0.36°. These results robustly support the hypothesis that gravitational deflection promotes greater angular deviation in the keyhole front wall. Additionally, flow field assessments within the molten pool's longitudinal section demonstrate a concurrent rise in maximum flow velocity with increasing θ, thereby corroborating the aforementioned conclusion.

The temperature fields and velocity fields at four different positions within the molten pool at the same instant under the keyhole coupling condition were studied as selected in Fig. 14. During horizontal DLBSW, the liquid metal that forms the bottom vortex flows downward along the liquid/solid interface and converges at the central position, leading to an impact and subsequent vertical flow toward the molten pool's upper part. A decrease in the z-component of gravity weakens the downward driving force on the vortex flow, while an increase in the y-component enhances the lateral flow of the vortex. This combination of changes induces a redirection of the symmetric bottom vortex flow before it reaches the bottom. As mentioned above, the disorderliness of the bottom vortex flow is intensified by the gravitational deflection, the energy and momentum of which result in the presence of a plurality of depressions within the keyhole coupling region that are in oscillation at all times.

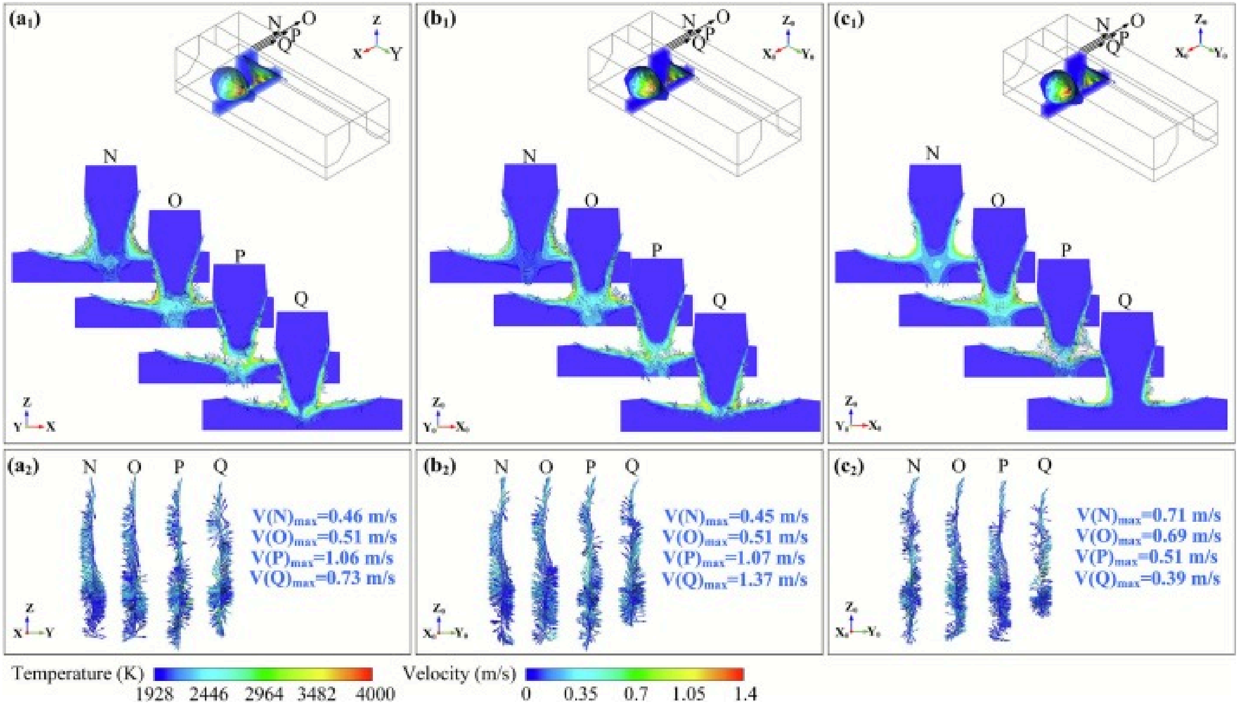

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Fig. 14. The temperature fields and velocity fields in cross sections of the coupling keyhole under different θ: (a) θ=0°; (b) θ=25°; (c) θ=40°.

Fig. 15 further depict the temperature fields and velocity fields in inclined sections of the coupling keyhole under different θ. Quantitative dimensions of the keyhole are extracted. During horizontal DLBSW, the average diameters of the keyhole opening region in the respective R, S, and T inclined sections are determined to be 2.254 mm, 1.609 mm, and 0.887 mm. At θ=25°, the average diameters of the keyhole opening region in the respective R, S, and T inclined sections are determined to be 2.296 mm, 1.607 mm, and 0.875 mm. At θ=40°, the average diameters of the keyhole opening region in the respective R, S, T inclined sections are determined to be 2.287 mm, 1.592 mm, and 0.914 mm. On the basis of the quantitative evaluations conducted, it can be drawn that the offset of the θ does not have a significant impact on the dimensions of the keyhole opening region. However, for the dimensional change of the keyhole coupling region, it was shown that the offset angle of gravity has a significant impact on it. In the case of a θ of 25°, the average diameter of the keyhole coupling region is significantly reduced to 0.256 mm and 0.115 mm, compared to 0.377 mm and 0.238 mm without gravity offset. There are two factors driving this transition. One is associated with the hindrance of the vapor jet transport to the coupling region by the bulges. The larger oblique angle of the keyhole's front wall and the greater number of bulges trap a large amount of the vapor jet in the opening region. Furthermore, the enhanced absorption of laser energy by the keyhole wall results in an increase in vapor flow velocity, which, in turn, intensifies the keyhole oscillation, further impeding the transport of vapor. Another important factor is the effectiveness of the laser energy delivered to the coupling region. The inflection in the connection between the coupling region and the opening region, along with the protrusions on the keyhole wall, results in a decrease in the direct and reflected light rays transmitted to the coupling region, causing lower temperatures and weaker recoil pressure.

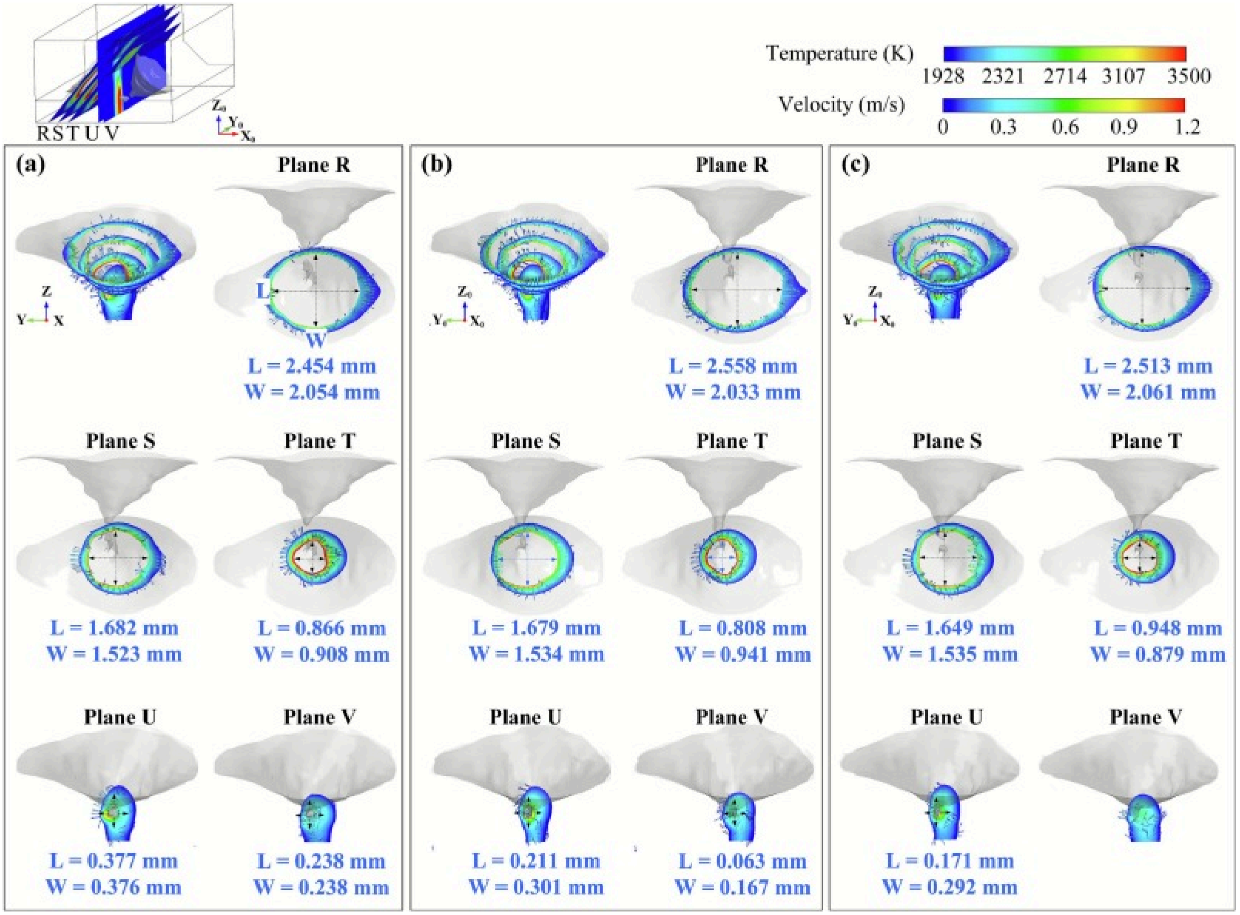

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Fig. 15. The temperature fields and velocity fields in inclined sections of the coupling keyhole under different θ: (a) θ=0°; (b) θ=25°; (c) θ=40°.

4.4. Bubble generation and porosity mitigation in DLBSW

It have demonstrated that the keyhole instability during DLBSW is intimately linked to the bubble generation. The suppression of keyhole oscillation and closure significantly mitigates the bubble formation, thereby enhancing process stability [33], [34]. Fig. 16

delineates the spatiotemporal evolution of the keyhole and the sequential emergence of bubbles. As evidenced in Fig. 16(a), the coupling region manifests a larger diameter and demonstrates greater dimensional stability than the transition region. Upon deflection of the coupling region from the laser irradiation axis, pronounced attenuation of incident laser energy ensues, elevating the surface tension of the keyhole wall and precipitating closure. Furthermore, vapor cooling induces a reduction in internal recoil pressure, thereby amplifying the disequilibrium between external forces and recoil pressure. As presented in Fig. 16(b), the coupling region experiences a central bulge due to liquid metal impingement, alongside a further decline in upper wall temperature. The flow dynamics in this region are predominantly horizontal, contrasting with the vertical flow observed in the opening region. The inflection point facilitates a marked redirection of liquid flow, thereby reinforcing the tendency toward keyhole closure and liquid bridge formation, as indicated in Fig. 16(c) and (d). During this phase, the lower section of the molten pool manifests a coherent flow structure, with a clockwise vortex on the right and a counterclockwise vortex on the left.

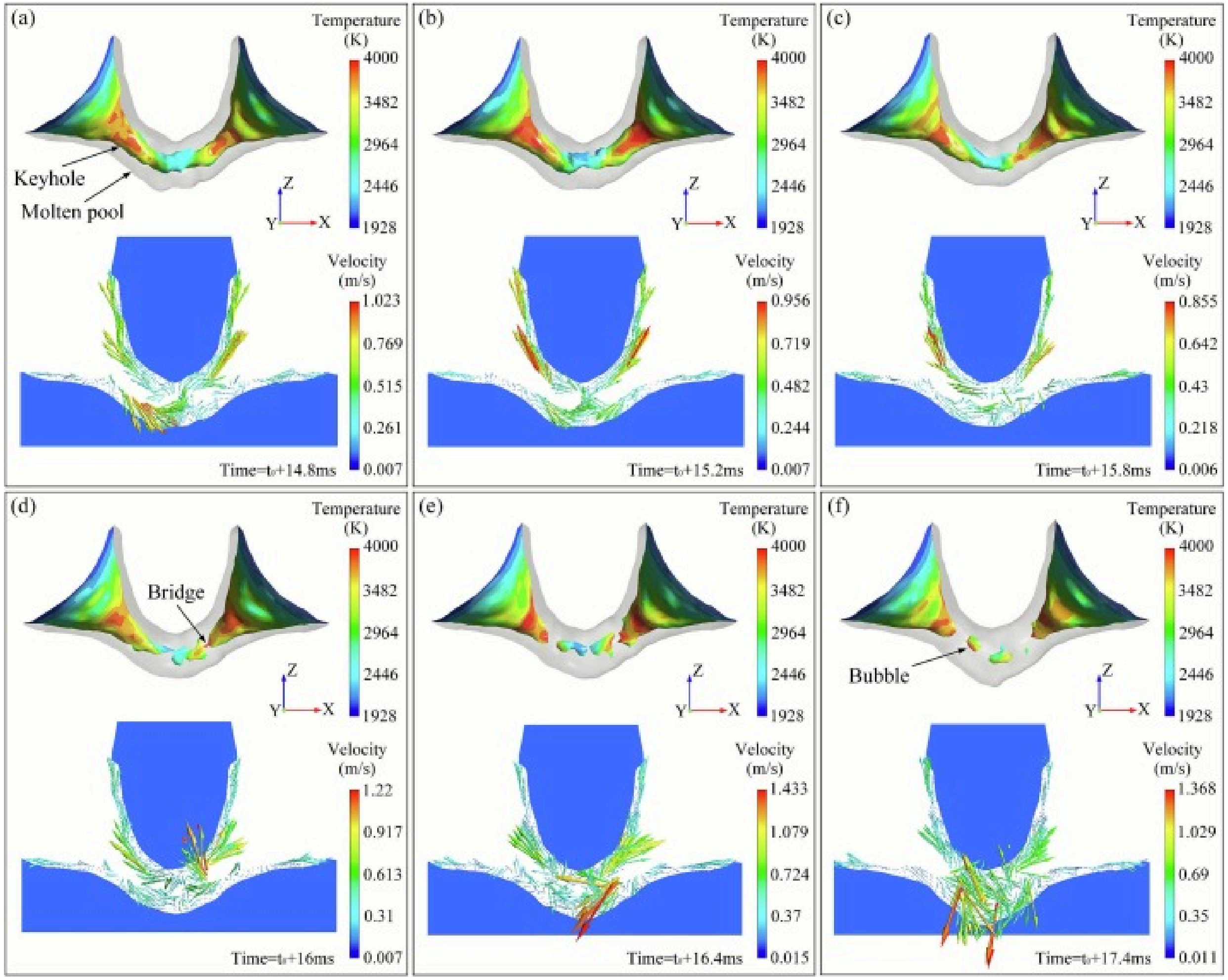

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Fig. 16. The dynamic evolution of the keyhole and the development of bubbles at various moments: (a) $t=t_0+14.8$ ms; (b) $t=t_0+15.2$ ms; (c) $t=t_0+15.8$ ms; (d) $t=t_0+16$ ms; (e) $t=t_0+16.4$ ms; (f) $t=t_0+17.4$ ms;

External liquid impingement may induce the formation of a closed or extremely constricted local zone within the coupling keyhole, where vigorous liquid convection prevails, facilitating primary fluid exchange between the upper and lower molten pool regions generated by keyhole incision, with flow velocities peaking at 1.433 m/s (Fig. 16(e)). Subsequent to the complete separation of the coupling keyhole and the emergence of an independent bubble, the flow field exhibits increased turbulence and a breakdown in symmetry and regularity. The heightened fluid velocity proximate to the keyhole constriction elevates dynamic pressure, which exacerbates keyhole oscillation and contraction. This dynamic suppresses the laser energy shielding effect attributable to internal keyhole topography, thereby promoting energy transfer and increasing the keyhole root temperature. The augmented thermal energy within the liquid reduces surface tension along the keyhole wall. Bubble formation is typically attributed to the entrapment of metal vapor during keyhole occlusion. The flow surrounding the bubble is largely vertical, oriented from the pool bottom to the surface. Given the T-shaped structural configuration, the bubble cannot escape via buoyant forces alone; instead, it is intercepted by the unmelted truss, leading to its immobilization within the pool and eventual incorporation into the solidification front as a gas pore. Therefore, lateral flow parallel to the skin is of paramount importance during bubble escape. A coherent flow regime diminishes the bubble's escape path, thereby improving egress efficiency and mitigating defect generation.

Fig. 17 elucidates the modulation of keyhole coupling frequency and initial coupling time by gravity deflection and welding speed. The coupling frequency diminishes as the θ escalates, a trend reminiscent of that provoked by diminished welding speed, though the mechanistic underpinnings diverge substantially. Specifically, increased gravity deflection destabilizes the coupling configuration, increasing the temporal dominance of separated or closed states. Under low welding speed regimes, the coupling frequency reduction stems from augmented expansion forces, which fortify the keyhole against liquid metal impact, thereby prolonging stable coupling and reducing oscillatory cycles. This prolongation of tip oscillation elevates the probability of bubble formation via keyhole rupture. As depicted in Fig. 17(b), the initial coupling time exhibits no coherent trend, indicating a lack of strong intrinsic association with welding speed or gravity deflection.

Furthermore, increased gravity deflection exacerbates flow heterogeneity, impeding bubble clearance and extending bubble residence in the coupling region.

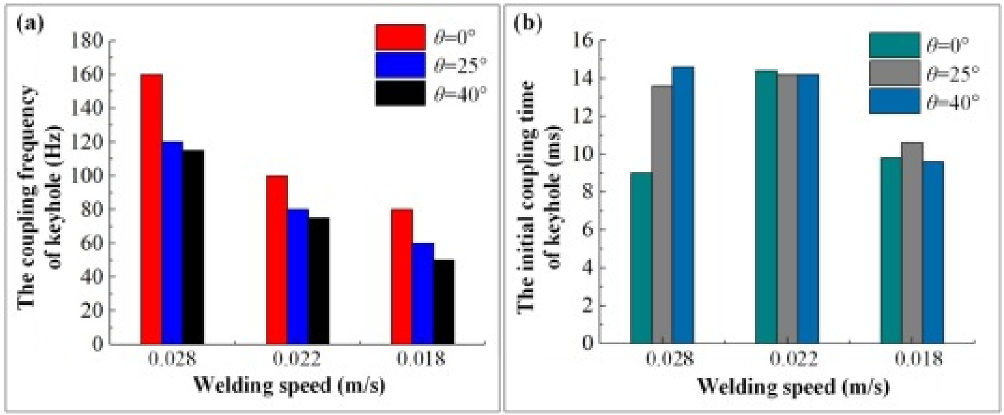

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Fig. 17. Comparison of keyhole stability under different gravity deflection states: (a) keyhole coupling frequency; (b) keyhole initial coupling time.

Fig. 18 delineates keyhole dynamic behavior during bubble generation under disparate gravity deflection regimes, informed by antecedent research on bubble evolution in dual laser beam welding. In the non-deflected state, the amalgamation of clockwise and counterclockwise vortices at the molten pool's nadir engenders a vertical liquid metal flux. Gravity deflection induces an oblique ascent trajectory, wherein the lateral velocity component augments bubble egress toward unconfined regions, attenuating porosity incidence. The upward shear stress resultant from Marangoni forces, surface tension, and thermal buoyancy operates concertedly to advect liquid along the keyhole's basal interface toward the pool's superior surface, whereas gravitational and counter-impulse pressures impose a downward vector. The gravitational deflection engenders a significant attenuation in the mass transport of liquid toward the molten pool's base, thereby curtailing the fluid's capacity to effectuate void occlusion through convective flow. The non-deflected molten pool, distinguished by reduced central flow velocity and attenuated impedance to unilateral keyhole expansion, exhibits superior bubble entrapment, thereby inhibiting porosity. The flow pattern analysis corroborates that gravity deflection profoundly modulates keyhole dynamics, perturbing liquid metal flow directionality and kinematics, which consequently impairs bubble clearance and elevates pore defect susceptibility.

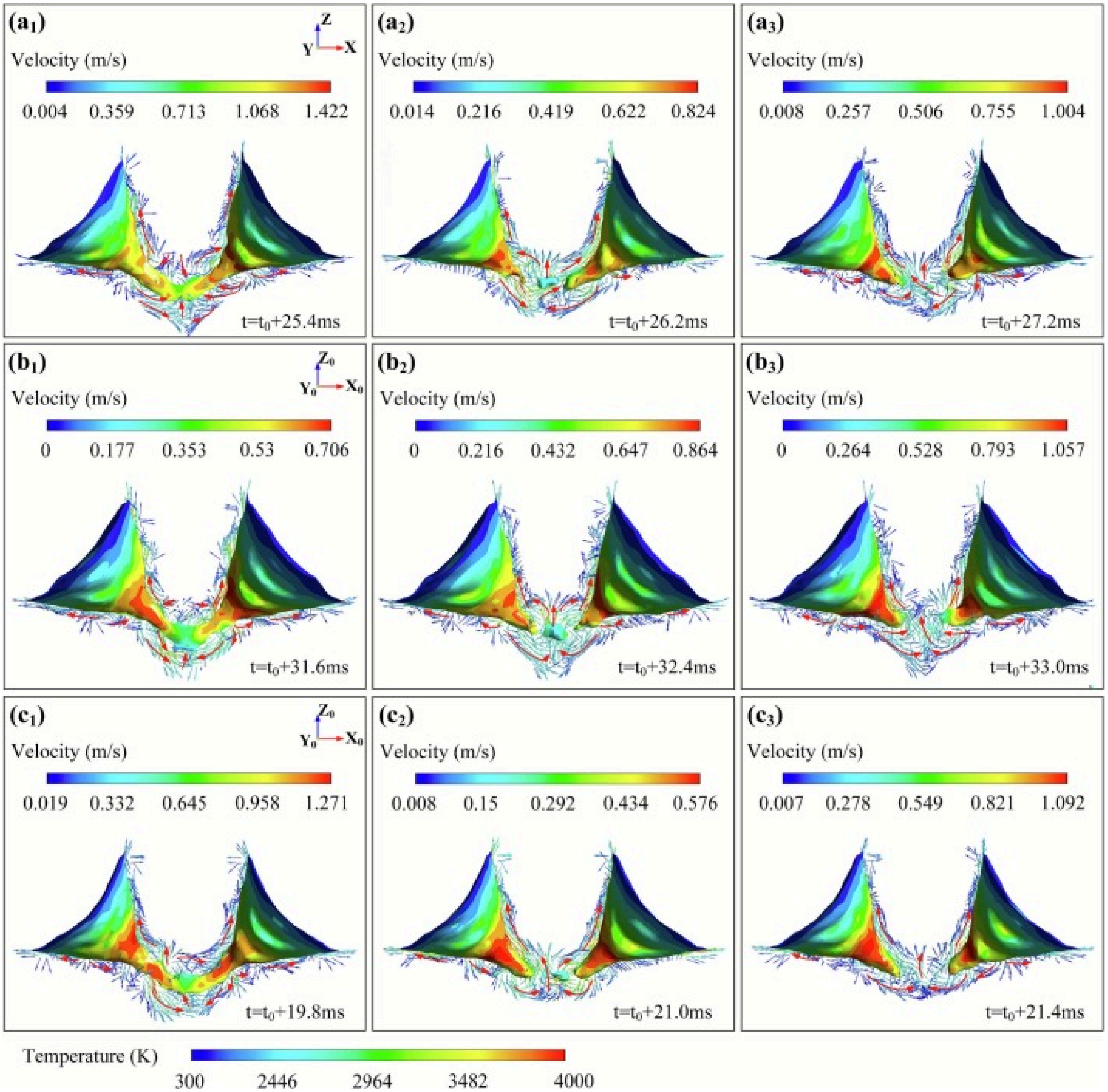

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Fig. 18. The dynamic behavior of the keyhole for the bubble generation process under various gravity deflection states: (a) θ=0°; (b) θ=25°; (c) θ=40°.

Fig. 19 presents a schematic of the keyhole induced-pore formation process under different gravity conditions in DLBSW. The occurrence of gravity deflection significantly shortens the size of molten pool flanks, which enhances the possibility of bubbles being captured by the solidification front and is not conducive to inhibiting the formation of porosity. Meanwhile, the clockwise vortex in the molten pool's bottom tends to trap bubbles at the bottom, hindering their upward mobilization. In accordance with the Stokes formula, the calculation of the bubble escaping speed is outlined as follows [35]:

$$V_e = \frac{2(\rho_m - \rho_g) g r^2}{9\eta_m} \tag{16}$$

where V_e represents the rate at which the bubble escapes from the surrounding liquid, ρ_m is the density of the liquid surrounding the bubble, ρ_g is the bubble density, η_m is the viscosity of the liquid, g and r are gravitational acceleration and bubble radius, respectively. The findings presented in this study indicate that gravity may affect the formation of porosity in multiple ways, for example, by decreasing the escape velocity of bubbles thereby further facilitating the formation of porosity. X-ray CT scanning was employed to ascertain the dimension and distribution of porosity within the weld, demonstrating the porosity at various speeds for titanium alloy T-joints welded at different θ. Fig. 20 illustrates that the porosity of the weld increases with gravity deflection, aligning with the analysis presented earlier. Specifically, the experimental data indicate that the porosity is minimized when there is no gravitational deflection and the velocity of welding is 0.018 m·s^{-1}.

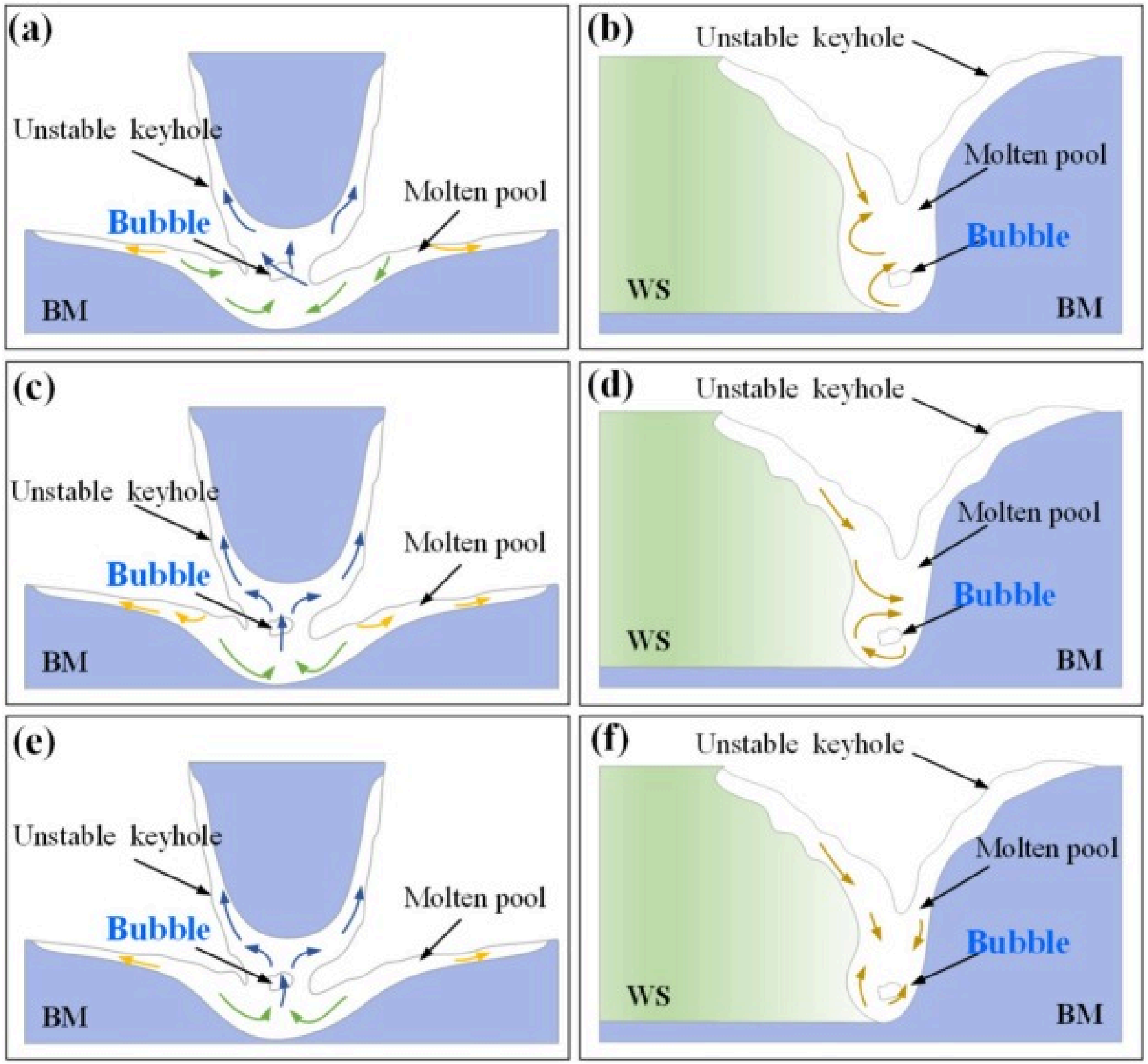

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Fig. 19. Flow patterns of the liquid metal in both the longitudinal section and cross-section of the molten pool: (a) θ=0°; (b) θ=25°; (c) θ=40°.

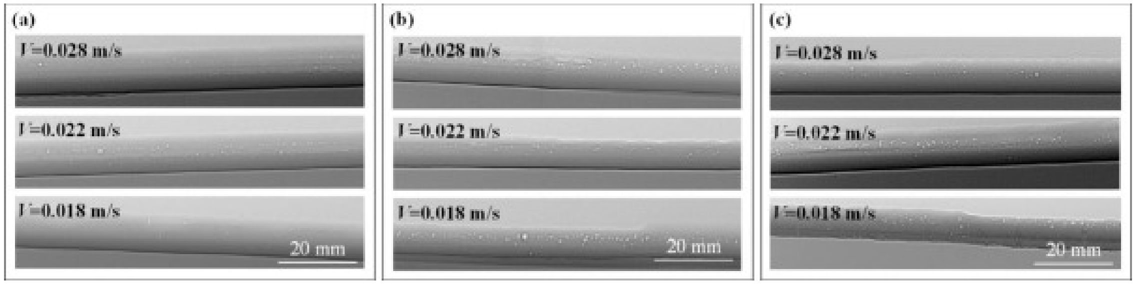

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Fig. 20. The porosity defects in the weld of T-joints under different gravitational deflection conditions: (a) θ=0°; (b) θ=25°; (c) θ=40°.

5. Conclusions

The present study employs computational simulations to analyze the volution mechanism of gravity-induced distortion dynamics for molten pool and keyhole during non-horizontal DLBSW of Ti-6Al-4V alloy T-joints. The following conclusions canbedrawn:

(1) Gravity deflection acts to increase the roughness of the weld surface by varying the magnitude of force components across directions. This facilitates a counterclockwise flow of the molten material along the rear wall of the keyhole, improving fluidity while simultaneously inducing localized reflux and intensifying fluctuations in the upper molten pool.

(2) Gravity deflection exhibits a pronounced correlation with the morphological smoothness and inclination of the keyhole front wall. As the θ increases, the flow pattern of the liquid metal in the upper region of the molten pool transitions from initially advancing toward the front wall to being redirected toward the rear wall and the central lower portion of the pool. This rerouting markedly diminishes the thermal energy transferred to the

anterior wall, thereby impeding the forward progression of the solid/liquid interface and exacerbating the non-uniformity of heat distribution.

(3) The inflection between coupling and opening regions, as well as protrusions on the keyhole wall, resulting in a reduction of direct and reflected light rays transmitted to the coupling region and traps a large amount of the vapor jet in the opening region. These changes lead to a lack of effective support on the keyhole wall from the inside. Therefore, reducing the θ and decreasing the θ is beneficial in suppressing oscillations and separations of the keyhole.

(4) This investigation elucidates the multifactorial role of gravity in porosity development, encompassing the promotion of bubble nucleation via keyhole collapse, the suppression of bubble transport, and the reduction of liquid metal's void-sealing capability. The dynamics of the molten pool and keyhole together with the gravitational impact mechanism were revealed.

CRediT authorship contribution statement

Yanqiu Zhao: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Tingyan Yan:** Writing – original draft, Investigation, Formal analysis. **Rui Li:** Validation, Methodology. **Jiangfeng Wang:** Investigation, Formal analysis. **Chao Ma:** Investigation. **Xiaohong Zhan:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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